

Altered reactivity to threatening stimuli in *Drosophila* models of Parkinson's disease, revealed by a trial-based assay

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eLife Assessment

The authors present **useful** findings on the use of a single-fly behavioral paradigm for assessing different *Drosophila* genetic models of neurodegeneration. The experimental design and analyses are **solid** and can be used for quick behavioral assessment in fly models of various neurodegenerative diseases, especially those having an impact on locomotion. The work will be of interest to *Drosophila* biologists using behavior as a readout for their studies.

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Abstract

The fruit fly *Drosophila melanogaster* emerges as an affordable, genetically tractable model of behavior and brain diseases. However, despite the surprising level of evolutionary conservation from flies to humans, significant genetic, circuit-level and behavioral differences hinder the interpretability of fruit fly models for human disease. Therefore, to allow a more direct fly-versus-human comparison, we surveyed the rarely-exploited, rich behavioral repertoire of fruit flies with genetic alterations relevant to Parkinson's disease (PD). Flies displayed variable behaviors, including freezing, slowing and running, in response to predator-mimicking passing shadows used as threatening stimuli in a single-animal trial-based assay. We found that the expression of human mutant Parkin in flies resulted in reduced walking speed and decreased reactivity to passing shadows. Flies with dopamine receptor mutations showed similar alterations, consistent with the motor and cognitive

deficits typical in humans with PD. We also found age-dependent trends in behavioral choice across the fly lifespan, while dopamine receptor mutant flies maintained their decreased general reactivity throughout all age groups. Our data demonstrate that single-trial behavioral analysis can reveal subtle behavioral changes in mutant flies that can be used to further our understanding of disease pathomechanisms and help gauge the validity of genetic *Drosophila* models of neurodegeneration, taking us one step closer to bridging the gap in fly-to-human translation.

Introduction

Parkinson's disease (PD) affects over 6 million people worldwide, representing the second most common neurodegenerative disease. PD is slowly progressing and typically leads to years of aggravating disability, thereby placing a huge burden on families, health care systems and the society, measured in hundreds of billions of dollars annually (Olesen et al., 2012; Obeso et al., 2017; Przedborski, 2017; Bloem et al., 2021). Therefore, a hitherto elusive disease-modifying therapy is of prominent priority, envisioned to be fueled by research into basic mechanisms of the disease.

Patients with PD display a diverse set of motor and non-motor symptoms with an underlying progressive loss of midbrain dopaminergic neurons (Schapira, 2009; Schapira et al., 2017; Bloem et al., 2021), mechanisms of which are widely studied in animal models of the disease. Primate models offer the advantage of more direct translatability but are restricted to hard-to-handle neurotoxin approaches and do not promise fast progress. Unlike the genetic mouse models of Alzheimer's disease, most of the successful rodent models of PD are also toxin-based, making it difficult to exploit and further our understanding of the genetic bases of the disease (Cannon and Greenamyre, 2010; Bové and Perier, 2012; Breger and Fuzzati Armentero, 2019). This left a niche for the genetically tractable, affordable fruit flies as genetic models of PD (Feany and Bender, 2000; Guo, 2012; Hewitt and Whitworth, 2017), building on the homologies between the vertebrate basal ganglia and the fruit fly central complex (Strausfeld and Hirth, 2013). However, substantial genetic, anatomical, physiological, and behavioral discrepancies between insects and mammals call for better validation methods of fruit fly models for human disease. Therefore, to provide means for a more direct comparison between flies and mammals, we developed a single-animal trial-based behavioral assay that facilitates fine-grained assessment of phenotypical behavioral changes in fruit flies with mutations relevant to understanding human PD.

Genes linked to familial forms of PD may serve as an ideal basis for genetic disease models. Notably, mutations in the *PARK2* gene, which encodes the Parkin protein involved in maintaining mitochondrial integrity, are associated with autosomal-recessive forms of PD (Guo, 2012). Parkin loss-of-function mutant flies were found to have advanced mitochondrial aging, structural mitochondrial damage and a consequential selective loss of dopaminergic neurons (Cackovic et al., 2018), leading to motor deficits assessed by a climbing assay (Chambers et al., 2013; Cackovic et al., 2018), as well as non-motor PD phenotypes including memory deficits (Julienne et al., 2017). Mutations in the *SNCA* gene of α -Synuclein (α -Syn) are also associated with familiar forms of PD (Polymeropoulos et al., 1997) and α -Syn has been shown to accumulate in Lewy-bodies and Lewy-neurites (Spillantini et al., 1997). α -Syn proteins have been found in the pre-synaptic terminals in humans and mice (Kahle et al., 2000) and are thought to be involved in regulating dopamine (DA) synthesis under physiological conditions by reducing tyrosine hydroxylase activity (Perez et al., 2002). Expression of human α -Syn in flies has been proposed as a genetic model of PD, showing age-dependent loss of dopaminergic neurons and locomotor dysfunction in a climbing assay (Feany and Bender, 2000; Haywood and Staveley, 2006). However, other studies found normal locomotion and dopaminergic cell counts in these flies, casting doubts on the validity of this model (Pesah et al., 2005; Nagoshi, 2018).

Despite the observed homology between mammalian and fruit fly DA systems in motor control and the establishment of *Drosophila* PD models based on human genetic information derived from familial PD patients, the role of *Drosophila* DA receptors in locomotor control is not well characterized. Nevertheless, it has been demonstrated that the D1-like DA receptor mediates ethanol-induced locomotion in the ellipsoid body (Kong et al., 2010 [↗](#)). Furthermore, *Dop1R1* has been shown to be involved in turning behavior for goal-directed locomotion (Kottler et al., 2019 [↗](#)) and startle-induced negative geotaxis (Sun et al., 2018 [↗](#)). To shed light on the possible roles of DA receptors in threat-induced motor behaviors, we tested the behavioral responses of three dopamine receptor (*Dop1R1*, *Dop1R2* and *DopEcR*) insertion mutant lines to predator-mimicking passing shadows and compared them to established fruit fly PD models with partially known locomotor deficits.

We found that flies expressing the R275W mutant allele of human Parkin ('Parkin flies') showed slower average locomotion speed, which, in contrast, was not characteristic of flies expressing the A53T mutant allele of the human *SNCA* gene (' α -Syn flies'). Parkin flies also showed less behavioral reactivity to passing shadows compared to controls, whereas α -Syn flies showed increased durations of stopping after the stimuli. Dopamine receptor mutant flies showed reduced speed and less behavioral reactivity similar to Parkin flies. *Dop1R1* mutant flies exhibited more pronounced behavioral alterations than the other two receptor mutants in most of the parameters tested. These data demonstrate that mutations in DA receptor genes lead to specific patterns of behavioral deficits in *Drosophila*; hence, these dopamine receptor paralogs may have different functions in behavioral control. The modest phenotype of A53T α -Syn compared to Parkin flies suggests that the latter should be favored as a genetic model of human PD, at least with respect to the motor deficits examined in this study. We further propose that single-trial analyses such as those we present here help us gain a better understanding of the behavioral changes in fruit fly models of PD and are strong tools for validating *Drosophila* models of human diseases.

Results

A single-animal trial-based assay to test behavioral responses to predator-mimicking passing shadows

We designed a behavioral apparatus to examine the responses of individual flies to predator-mimicking passing shadows. To do this, we designed a transparent plexiglass arena (**Fig. 1a** [↗](#)), featuring 13 tunnels (53 mm x 5 mm) for simultaneous tracking of 13 individual flies. The height of the tunnels (1.5 mm) was designed to allow free walking in two dimensions but prevent jumps and flight. To simulate predatory threat, we created passing 'shadows' (**Fig. 1a** [↗](#), right; see Methods) with a sliding red screen presented on a 10.1-inch display placed on the top of the arena. A high frame-rate camera was placed under the arena (**Fig. 1a** [↗](#), left), allowing us to simultaneously record the movement of the animals and the shadows (Supplementary Movie 1 and 2). All 13 tunnels housed a single fly in each session, enabling the collection of single-animal data from 13 flies in parallel. The locomotion of individual flies was tracked by custom software developed using the Bonsai visual programming environment (**Fig. S1** [↗](#), see Methods). We recorded 40-minute-long sessions, consisting of 40 trials of 2-second-long shadow presentations separated by pseudorandom inter-trial-intervals to prevent the animals from learning temporal expectations of the shadow presentations (**Fig. 1b** [↗](#)).

Fruit flies exhibit a rich behavioral repertoire upon threatening stimuli, including freezing (or stopping) and various escape behaviors such as jumping, slow or fast take-off and running, modulated by walking speed at the time of the threatening stimulus (Zacarias et al., 2018 [↗](#)). To study these behaviors, we calculated the speed and acceleration of fruit flies based on the tracked x-y position of their center of mass (**Fig. 2a** [↗](#)) and aligned these signals to the presentations of the shadow stimuli. To identify separate response types to the threatening stimuli, we applied

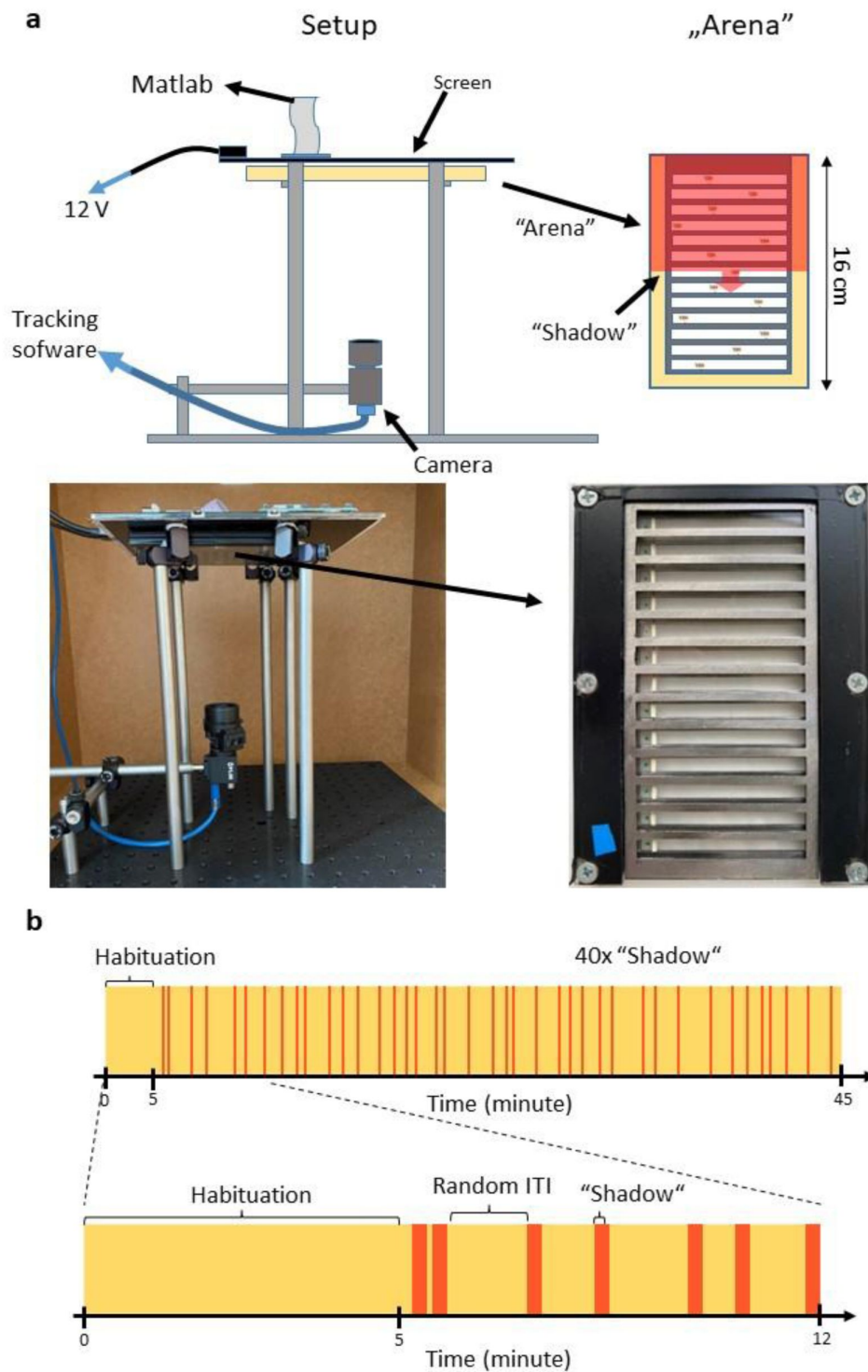


Fig. 1.

Behavioral apparatus and experimental design.

a, Top left, schematic of the experimental setup. Top right, schematic of the fly arena. Bottom left, photograph of the experimental setup. Bottom right, a photograph of the fly arena. **b**, Timeline of a session showing the shadow presentations in red over the yellow background.

hierarchical clustering on these stimulus-aligned speed traces (see Methods). This analysis consistently revealed three main stereotypical behavioral responses across flies in addition to the trials where no significant response was evoked (**Fig. 2b**). Since jumping and flying was not possible in the arena, fruit flies were restricted to choose among freezing, slowing (also observed in (Zacarias et al., 2018)) and running. While this clustering approach was sufficient to reveal response types qualitatively, cluster boundaries were sensitive to genotype-specific differences across groups of flies (Király and Hangya, 2022). Therefore, for rigorous group comparisons, we determined exact definitions of each response type based on fly speed and acceleration (**Fig. 2c**). Reactions were considered robust if the absolute value of the acceleration reached 200 mm/s^2 ; otherwise, the trial was classified as a ‘no reaction’ trial. We considered the reaction as a ‘stop’ if the animal was moving before shadow presentation and its speed decreased to zero in the first second relative to the shadow presentation. If fly speed decreased to a non-zero value, the trial was defined as ‘slow down’, and if the fly accelerated after the shadow presentation, the trial was classified as ‘speed up’ (see also Methods).

Parkinson’s model and dopamine receptor mutant fruit flies showed reduced walking speed and decreased reactivity to threatening stimuli

To generate PD fly models, we expressed the mutant human Parkin (275W) and α -Syn (A53T) coding transgenes (*UAS-Parkin-275W* and *UAS- α -Syn-A53T*, respectively), applying the UAS-Gal4 system (Duffy, 2002). Transgenes were driven by *Ddc-Gal4* inducing the expression of the correspondent human mutant proteins in dopaminergic and serotonergic fly neurons. Parkin (275W) and α -Syn (A53T) flies were compared to control animals from the same genetic background without mutant transgenes (*Ddc-Gal4* females were crossed with *isogenic w¹¹¹⁸* males; the examined F1 generation flies are referred to as *iso w¹¹¹⁸*, **Fig. 3**) or mutants overexpressing GFP (**Fig. S2**). The same level of eye pigmentation and vision of the compared genotypes was achieved by the prior replacement of the *w** mutant X chromosome of the applied *Ddc-Gal4* stock for that of the wild-type. We also used Mi{MIC} random insertion lines for dopamine receptor mutants, namely *y¹ w*; Mi{MIC}Dop1R1^{MI04437}* (BDSC 43773), *y¹ w*; Mi{MIC}Dop1R2^{MI08664}* (BDSC 51098) (Pimentel et al., 2016; Harbison et al., 2019), and *w¹¹¹⁸; PBac{PB}DopEcR^{c02142}/TM6B, Tb¹* (BDSC 10847) (Ishimoto et al., 2013; Petrucci et al., 2016, 2020), referred to as Dop1R1, Dop1R2 and DopEcR, respectively. The dopamine receptor mutant groups were compared to their parental strains without the mutations (*y¹ w^{67c23}* served as control for *Dop1R1* and *Dop1R2* and *w¹¹¹⁸* for *DopEcR*; **Fig. 3**).

To test whether mutant flies showed differences in overall locomotion independent of the threatening stimuli, we analyzed average fly speed in the 200 ms time windows before stimulus presentation. We found that Parkin flies showed reduced mean speed compared to controls which expressed the Gal4 driver alone (24.96% reduction, $p = 6.55 \times 10^{-6}$, Mann-Whitney U-test; **Fig. 3a**, top), while α -Syn flies did not show significant reduction ($p = 0.799$, Mann-Whitney U-test). We observed similar changes in dopamine receptor mutant flies, where the *Dop1R1* and *DopEcR* defective lines showed a robust mean speed decrease compared to the control groups (*Dop1R1*, 19.68% decrease compared to *y¹ w^{67c2}*, $p = 0.0016$; *DopEcR*, 32.97% decrease compared to *w¹¹¹⁸*, $p = 0.0034$; **Fig. 3a**, bottom), while the *Dop1R2* mutant flies only showed a non-significant speed decrease (21.73% compared to *y¹ w^{67c23}*, $p = 0.1009$, Mann-Whitney U-test).

Next, we tested whether mutant flies showed a difference in their reaction to threatening stimuli. We found that PD model flies showed line-specific alterations in their freezing behavior. Parkin flies froze after stimuli less frequently, stopping in 37.72% of trials compared to the 43.48% observed in the *iso w¹¹¹⁸* animals ($p = 0.0043$, Mann-Whitney U-test; **Fig. 3b**, top). However, in the stop trials, they showed normal duration of pauses in locomotion ($p = 0.8018$ compared to *iso w¹¹¹⁸*, Mann-Whitney U-test; **Fig. 3c**, top). In contrast, α -Syn flies showed an unchanged

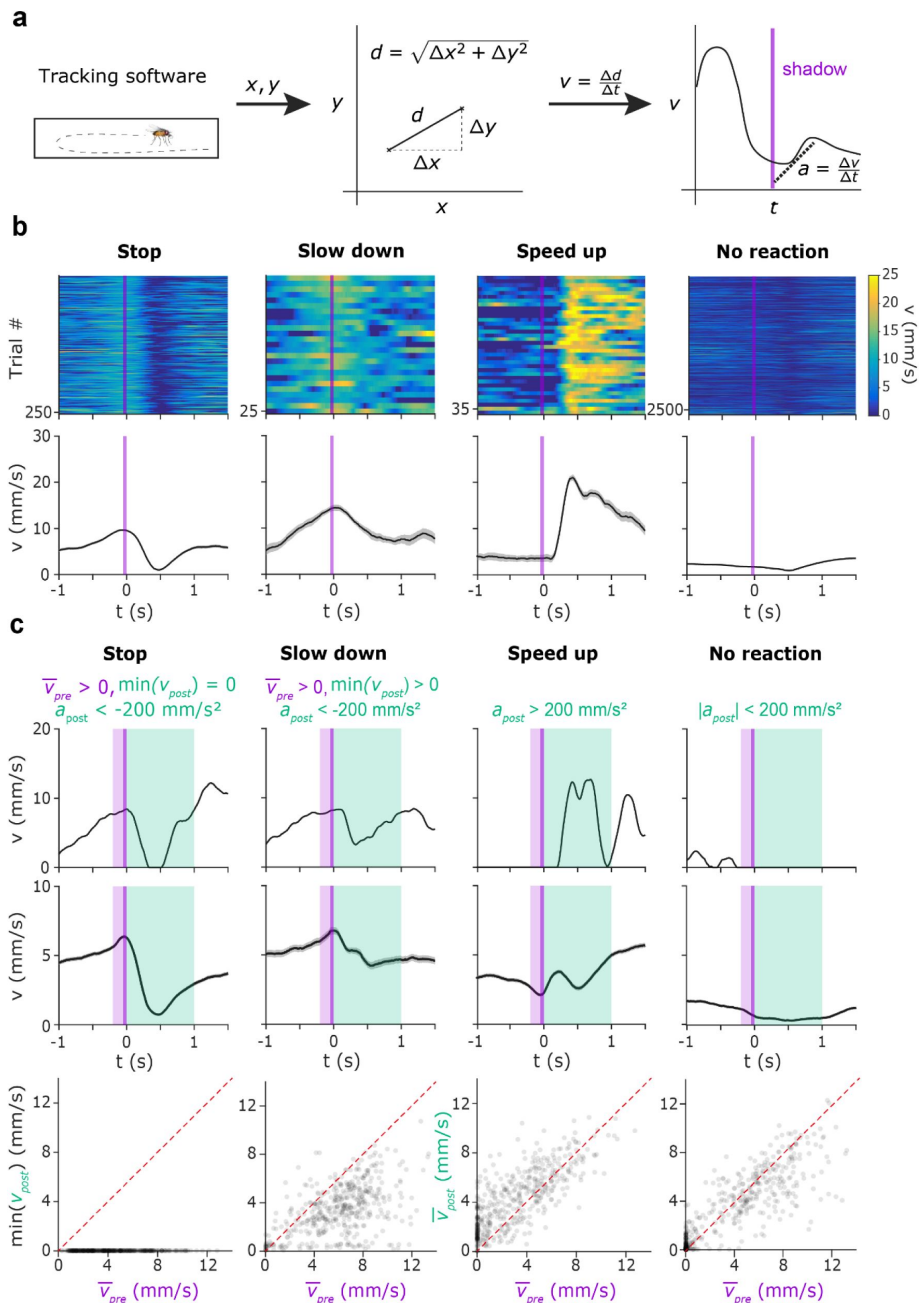


Fig. 2.

Behavioral characterization of the escape behavior repertoire of individual fruit flies.

a, Schematic for calculating speed and acceleration based on tracked position coordinates. **b**, Four characteristic escape behaviors categorized by PCA for an example session of a control fly (w¹¹¹⁸; from left; stop, slow down, speed up, no reaction). Top, color-coded heatmaps indicating the walking speed of the fly (blue, low speed; yellow, high speed), aligned to shadow presentations (purple line). Bottom, average moving speed triggered on the shadow presentations (purple line). **c**, Threshold-based classification of behavioral responses (from left, stop, slow down, speed up, no reaction). Top, single-trial example for each response type. Time intervals for calculating the average speed before the shadow presentation (purple) as well as the speed and acceleration after the shadow presentation (light green) are marked. Threshold values for each response type are displayed above the graphs. Middle, average walking speed across the trials from all sessions of w¹¹¹⁸ flies, sorted by the type of behavioral response. Line and errorshade show mean $\pm$ SE. Bottom, scatter plots showing minimal (Stop and Slow down trials) or average (Speed up and No reaction trials) post-stimulus speed vs. average pre-stimulus speed, representing the same trials as above.

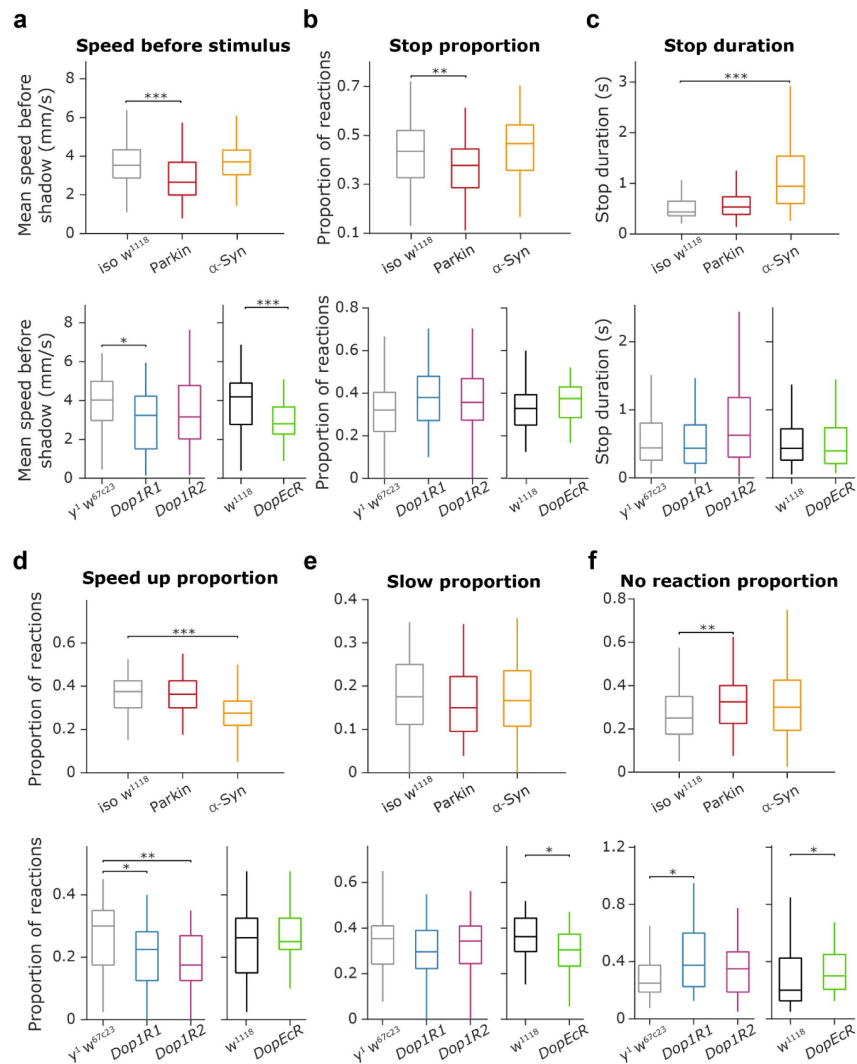


Fig. 3.

Parkinson's model and dopamine receptor mutant fruit flies showed reduced walking speed and decreased reactivity to threatening stimuli.

a, Distribution of the mean speed measured in the time window $[-0.2, 0]$ seconds relative to shadow presentation for the different mutant groups. Top, Parkin flies showed reduced mean speed compared to controls (Parkin vs. *iso w¹¹¹⁸*; $p = 6.55 \times 10^{-6}$, Mann-Whitney U-test). Bottom, *Dop1R1* and *DopEcR* mutant flies showed reduced mean speed compared to controls (*y¹ w^{67c23}* and *w¹¹¹⁸* respectively; $p = 0.0016$, $p = 0.0034$; Mann-Whitney U-test). **b**, Distribution of the proportion of stop trials in different mutant groups. Parkin flies showed a reduced tendency for stopping compared to *iso w¹¹¹⁸* ($p = 0.0043$, Mann-Whitney U-test). **c**, Distribution of stop duration in different mutant groups. α-Syn flies showed increased stop durations compared to *iso w¹¹¹⁸* ($p = 2.18 \times 10^{-11}$, Mann-Whitney U-test). **d**, Distribution of the proportion of speed up trials in different mutant groups. Top, α-Syn flies showed a reduced tendency to speed up compared to their controls ($p = 8.4 \times 10^{-6}$, Mann-Whitney U-test). Bottom, *Dop1R1* and *Dop1R2* mutant flies also showed a significantly reduced tendency to speed up compared to their controls ($p = 0.0122$ and $p = 0.0024$, respectively; Mann-Whitney U-test). **e**, Distribution of the proportion of slow down trials in different mutant groups. Bottom, *DopEcR* showed a 15.04% decrease compared to *w¹¹¹⁸* ($p = 0.020$, Mann-Whitney U-test). **f**, Distribution of the proportion of 'no reaction' trials in different mutant groups. Top, Parkin mutants showed reduced reactivity compared to *iso w¹¹¹⁸* controls ($p = 0.0043$, Mann-Whitney U-test). Bottom, *Dop1R1* and *DopEcR* mutants also showed reduced reactivity compared to *y¹ w^{67c23}* and *w¹¹¹⁸* controls, respectively ($p = 0.0173$ and $p = 0.0167$, respectively; Mann-Whitney U-test). Box-whisker plots show median, interquartile range and non-outlier range. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. Exact genotypes: *iso w¹¹¹⁸*: +; +; *Ddc-Gal4*/+. Parkin: +; +; *Ddc-Gal4/UAS-Parkin-R275W*. α-Syn: +; +; *Ddc-Gal4/UAS-α-Syn-A53T*. *y¹ w^{67c23}*: *y¹ w^{67c23}*. *Dop1R1*: *y¹ w^{67c23}*; *MI{MIC}Dop1R1^{MI04437}*. *Dop1R2*: *y¹ w^{67c23}*; *MI{MIC}Dop1R2^{MI08664}*. *DopEcR*: *w¹¹¹⁸*; *PBac{PB}DopEcR^{C02142}/TM6B*, *Tb¹*.

stopping frequency when compared to controls ($p = 0.0768$ compared to *iso w¹¹¹⁸*, Mann-Whitney U-test), but their stop durations showed a large increase (116.98% increase compared to *iso w¹¹¹⁸*, $p = 2.18 \times 10^{-11}$, Mann-Whitney U-test). Interestingly, in contrast to PD model flies, dopamine receptor mutant flies did not show significant differences in their freezing behavior relative to controls (stop proportion: *Dop1R1*, $p = 0.0631$ compared to *y¹ w^{67c23}*; *Dop1R2*, $p = 0.1838$ compared to *y¹ w^{67c23}*; *DopEcR*, $p = 0.1445$ compared to *w¹¹¹⁸*, Mann-Whitney U-test; **Fig. 3b**, bottom; stop duration: *Dop1R1*, $p = 0.893$ compared to *y¹ w^{67c23}*; *Dop1R2*, $p = 0.154$ compared to *y¹ w^{67c23}*; *DopEcR*, $p = 0.3576$ compared to *w¹¹¹⁸*, Mann-Whitney U-test; **Fig. 3c**, bottom).

α -Syn flies showed a reduced aptitude to increase their speed, or ‘run’, upon encountering threatening stimuli (26.67% decrease compared to *iso w¹¹¹⁸*, $p = 8.4 \times 10^{-6}$, Mann-Whitney U-test; **Fig. 3d**, top). Similar results were found in *Dop1R1* (25% decrease compared to *y¹ w^{67c23}*, $p = 0.0122$; Mann-Whitney U-test; **Fig. 3d**, bottom) and *Dop1R2* (41.67% decrease compared to *y¹ w^{67c23}*, $p = 0.0024$, Mann-Whitney U-test), but not in *DopEcR* mutant flies (compared to *w¹¹¹⁸*, $p = 0.6598$, Mann-Whitney U-test). Frequency of slowing, i.e., reducing their speed without freezing in a full stop, was moderately decreased in Parkin (14.38% decrease compared to *iso w¹¹¹⁸*, $p = 0.0723$, Mann-Whitney U-test; **Fig. 3e**, top) and *DopEcR* mutant flies (15.28% decrease compared to *iso w¹¹¹⁸*, $p = 0.020$, Mann-Whitney U-test; **Fig. 3e**, bottom).

Overall, all mutations tested resulted in decreased reactivity to threatening stimuli, confirmed by a significant increase in the proportion of trials where no reactions were detected (**Fig. 3f**). This effect was significant for Parkin flies (30% increase compared to *iso w¹¹¹⁸*, $p = 0.0043$, Mann-Whitney U-test), as well as for *Dop1R1* (50% increase compared to *y¹ w^{67c23}*, $p = 0.0173$, Mann-Whitney U-test) and *DopEcR* mutant flies (50% increase compared to *iso w¹¹¹⁸*, $p = 0.0167$, Mann-Whitney U-test), but not for α -Syn (α -Syn vs. *iso w¹¹¹⁸*, $p = 0.151$, Mann-Whitney U-test) and *Dop1R2* mutant flies (*Dop1R2* vs. *y¹ w^{67c23}*, $p = 0.1903$, Mann-Whitney U-test).

While *Ddc-Gal4* lines are widely used (Wang et al., 2007), they express mutations both in dopaminergic and serotonergic neurons, preventing the assessment of whether the behavioral differences were due to one of those cell types alone. To address this, we tested *NP6510-Gal4-R275W* (Parkin) and *NP6510-Gal4-A53T* (α -Syn) as well as *TH-Gal4-R275W* (Parkin) and *TH-Gal4-A53T* (α -Syn) flies where expression of the mutation was restricted to dopaminergic cells (Riemensperger et al., 2013). Results from these tests were largely similar to those of the *Ddc-Gal4* lines (**Fig. S3**), reproducing the decreased speed (*NP6510-Gal4-R275W*, 30.3% decrease compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 1.55 \times 10^{-6}$; *NP6510-Gal4-A53T*, 20.4% decrease compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 5.82 \times 10^{-10}$; *TH-Gal4-R275W*, 13.08% decrease compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 1.49 \times 10^{-6}$; *TH-Gal4-A53T*, 17.57% decrease compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 6.75 \times 10^{-7}$; Mann-Whitney U-test) and decreased overall reactivity (*NP6510-Gal4-R275W*, 57.14% increase in ‘No reaction’ compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 3.49 \times 10^{-8}$; *NP6510-Gal4-A53T*, 14.28% increase in ‘No reaction’ compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 0.1658$; *TH-Gal4-R275W*, 66.7% increase in ‘No reaction’ compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 2.77 \times 10^{-10}$; *TH-Gal4-A53T*, 66.7% increase in ‘No reaction’ compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 1.11 \times 10^{-6}$; Mann-Whitney U-test) of PD-model flies. Nevertheless, *TH-Gal4* and *NP6510-Gal4* mutants showed an increased propensity to stop (*NP6510-Gal4-R275W*, 16.47% increase compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 4.4 \times 10^{-11}$; *NP6510-Gal4-A53T*, 20.17% increase compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 1.38 \times 10^{-17}$; *TH-Gal4-R275W*, 16.32% increase compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 1.55 \times 10^{-4}$; *TH-Gal4-A53T*, 22.41% increase compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 8.82 \times 10^{-10}$, Mann-Whitney U-test). Stop duration showed a significant increase not only in α -Syn but also in Parkin fruit flies (*NP6510-Gal4-R275W*, 42.6% increase compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 5.99 \times 10^{-5}$; *NP6510-Gal4-A53T*, 69.5% increase compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 4.33 \times 10^{-11}$; *TH-Gal4-R275W*, 33.62% increase compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 1.63 \times 10^{-6}$; *TH-Gal4-A53T*, 37.48% increase compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 3.89 \times 10^{-7}$; Mann-Whitney U-test). Overall, this suggests that mutations in the dopaminergic neurons alone were sufficient to produce the above-described phenotypical changes.

After these detailed comparisons of central tendencies of behavioral data, we also tested whether behavioral variability within fly response types showed differences across mutant groups. We found that stop duration showed a significantly larger dispersion in α -Syn flies compared to controls ($p = 0.0001$, permutation test; **Fig. S4**), while no other significant difference in data variance was detected.

Reaction to threatening stimuli depends on fly walking speed

It has been shown that fruit flies may exhibit a different choice of escape behavior based on their momentary speed when encountering the threat (Zacarias et al., 2018). Therefore, we tested whether mutations in genes relevant to PD caused a change in this speed - behavioral response relationship. We calculated the probability of stopping, slowing, speeding up and no reaction as a function of speed for all mutants, as well as the ratio of each response type conditioned on walking speed at the time of shadow presentations (**Fig. 4a-b**). These analyses confirmed that ‘running’ and ‘no reaction’ was most likely at slow (< 5 mm/s) or zero walking speeds, while slowing down was more frequent as the walking speed increased. Freezing was most frequently observed in the 5-13 mm/s speed range. We did not observe significant difference between groups in their speed - behavioral response relationship for any of the response types (stops, $p = 0.4226$; slow down, $p = 0.6025$; speed up, $p = 0.9074$; no reaction $p = 0.4692$; two-way ANOVA genotype $\times$ speed interaction).

Since the distribution of the expression of various escape behaviors depended on walking speed, which was also different across the mutant lines tested (**Fig. 3**), we asked whether the difference in baseline speed could explain the observed differences in behavioral reactions. Therefore, we performed simulations where reaction probabilities were based on the baseline walking speed distribution of each mutant line to test whether behavioral responses could be predicted based on speed alone (**Fig. S5**). In most of the groups we found a significant difference between the predicted and the measured response type distributions, suggesting a role for other behavioral differences among the mutant lines beyond the baseline speed differences (Parkin, $p = 0.0487$; α -Syn, $p < 0.000001$; *Dop1R1*, $p = 0.332$; *Dop1R2*, $p = 0.006157$; *DopEcR*, $p = 0.003901$, chi-square test).

Changes in escape behavior from the first to the fourth week of life

The age of *Drosophila* may have a significant influence on their responses to threatening stimuli. To test this, we examined control and dopamine receptor mutant flies (w^{1118} , $y^1 w^{67c23}$, *Dop1R1*, *Dop1R2*, *DopEcR*) in 5 age groups: from 1-day-old to 4-week-old (**Fig. 5**; **Fig. S6**). The animals were kept under conditions that accelerated their aging (29°C and 70% humidity). Our data showed that the behavior of 1-day-old flies was significantly different from all other age groups. One-day-old flies stopped significantly more often, and the frequency of stopping gradually decreased with age in all groups (two-way ANOVA, age, $p = 1.87 \times 10^{-20}$, $f = 26.11$; genotype, $p = 1.3 \times 10^{-29}$, $f = 38.42$; genotype $\times$ age, $p = 3.18 \times 10^{-6}$, $f = 3.54$; 1-day-old vs. 1-week-old, $p = 1.1 \times 10^{-4}$; 1-week-old vs. 2-week-old, $p = 0.0074$; 2-week-old vs. 3-week-old, $p = 0.1015$; 3-week-old vs. 4-week-old, $p = 0.0079$, Mann-Whitney test for post hoc analysis). In contrast, the probability of slowing down showed a substantial increase from 1-day-old to 1-week-old animals, and then remained relatively stable until the 4th week (two-way ANOVA, age, $p = 9.11 \times 10^{-15}$, $f = 18.72$; genotype, $p = 1.4 \times 10^{-21}$, $f = 27.59$; genotype $\times$ age, $p = 0.0028$, $f = 2.291$; 1-day-old vs. 1-week-old, $p = 5.9 \times 10^{-10}$; 1-week-old vs. 2-week-old, $p = 0.992$; 2-week-old vs. 3-week-old, $p = 0.0305$; 3-week-old vs. 4-week-old, $p = 0.033$; Mann-Whitney test for post hoc analysis). The probability of speeding up showed a similar trend, except that the 4-week-old animals showed a significant decrease compared to 3-week-old flies (two-way ANOVA, age, $p = 2.48 \times 10^{-15}$, $f = 19.44$; genotype, $p = 2.28 \times 10^{-44}$, $f = 59.48$; genotype $\times$ age, $p = 1.1 \times 10^{-7}$, $f = 4.12$; 1-day-old vs 1-week-old, $p = 2.64 \times 10^{-10}$; 1-week-old vs. 2-week-old, $p = 0.572$; 2-week-old vs. 3-week-old, $p = 0.3445$; 3-week-old vs. 4-week-old, $p = 0.0179$;

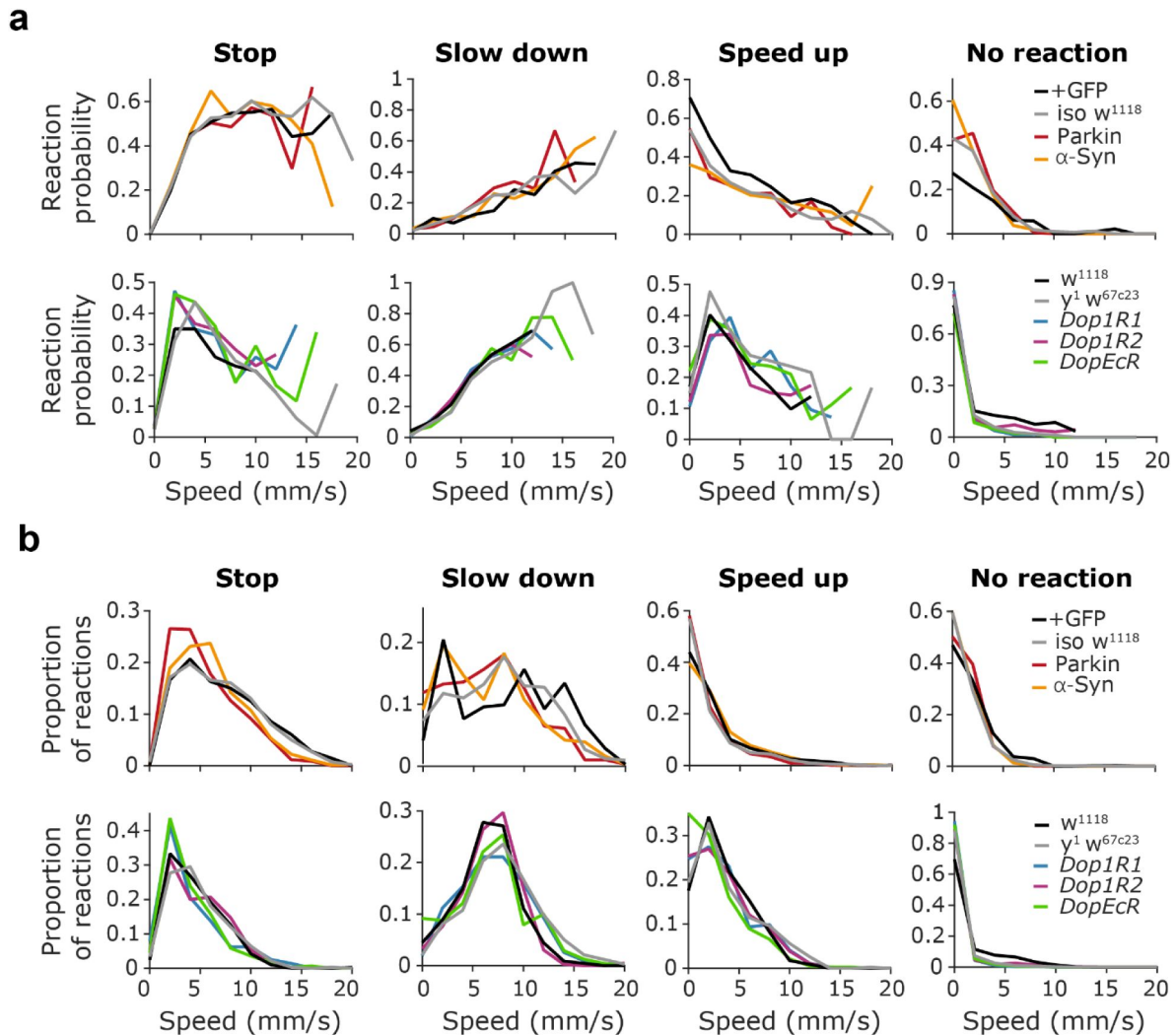


Fig. 4.

Reaction to threatening stimuli depends on fly walking speed.

a, Probability of a given response type as a function of average fly speed before the shadow presentation (200 ms pre-stimulus time window). From left, stop, slow down, speed up and no reaction trials are quantified. Top, Parkin and α -Syn flies. Bottom, *Dop1R1*, *Dop1R2* and *DopEcR* mutant flies. **b**, Proportion of a given response type as a function of average fly speed before the shadow presentation. Top, Parkin and α -Syn flies. Bottom, *Dop1R1*, *Dop1R2* and *DopEcR* mutant flies. Exact genotypes: *iso w¹¹¹⁸*: +; +; *Ddc-Gal4/+*. *Parkin*: +; +; *Ddc-Gal4/UAS-Parkin-R275W*. α -Syn: +; +; *Ddc-Gal4/UAS- α -Syn-A53T*. $y^1 w^{67c23}$: $y^1 w^{67c23}$. *Dop1R1*: $y^1 w^*$; *MI(MIC)Dop1R1^{MI04437}*. *Dop1R2*: $y^1 w^*$; *MI(MIC)Dop1R2^{MI08664}*. *DopEcR*: w^{1118} ; *PBac(PB)DopEcR^{C02142}/TM6B, Tb¹*.

Mann-Whitney test for post hoc analysis). Thus, different types of escape reactions showed remarkable tendencies across the fly lifespan, while dopamine receptor mutant flies maintained their decreased general reactivity throughout all age groups.

Discussion

We tested *Drosophila* mutant strains relevant for Parkinson's disease in a single-trial single-animal behavioral assay. Our tests revealed strain-specific behavioral alterations in flies' reactions to predator-mimicking passing shadows, serving as proof of principle demonstration of the viability of single-trial approaches in *Drosophila*, a method widely used in mammalian studies. Specifically, we found reduced walking speed, decreased freezing frequency and decreased overall reactivity in Parkin flies. In contrast, α -Syn flies merely showed an increased freezing duration without a concomitant change in freezing frequency, suggesting that Parkin flies better recapitulate some behavioral features of human PD progression. *Dop1R* mutant flies resembled Parkin flies in their decreased walking speed and decreased reactivity to the threatening stimuli. The distribution of the behavioral response fruit flies chose to execute depended on their speed at the time of stimuli, and this speed - behavioral response relationship was robust across the tested genotypes. Nevertheless, differences in walking speed alone did not explain the strain-specific behavioral alterations. Age-dependence of behavioral choice was also found to be stereotypic across the tested genotypes.

Recent studies increasingly recognize the importance of investigating subtle changes in fly behavior to better understand the manifestations of locomotor and other disorders (Geissmann et al., 2017 [DOI](#); Zacarias et al., 2018 [DOI](#); Seidenbecher et al., 2020 [DOI](#)). Along these lines, Aggarwal and colleagues introduced an automated climbing assay for fruit flies and revealed subtle motor deficits in different mutant strains (Aggarwal et al., 2019 [DOI](#)). We expanded the scope of these studies by examining a diverse behavioral repertoire of *Drosophila* in response to threatening stimuli. Many earlier studies used looming stimulus, that is, a concentrically expanding shadow, mimicking the approach of a predator from above, to study escape responses in flies (Card and Dickinson, 2008 [DOI](#); de Vries and Clandinin, 2012 [DOI](#); Zacarias et al., 2018 [DOI](#); Ache et al., 2019 [DOI](#); Oram and Card, 2022 [DOI](#)) as well as rodents (Lecca et al., 2017 [DOI](#); Braine and Georges, 2023 [DOI](#); Heinemans and Moita, 2024 [DOI](#)). These assays have the advantage of closely resembling naturalistic, ecologically relevant threat-inducing stimuli, and allow a relatively complete characterization of the fly escape behavior repertoire (Zacarias et al., 2018 [DOI](#)). As a flip side of their large degree of freedom, they do not lend themselves easily to provide a fully standardized, scalable behavioral assay. Therefore, Gibson et al. suggested a novel threat-inducing assay operating with moving overhead translational stimuli, that is, passing shadows, and demonstrated that they induce escape behaviors in flies akin to looming discs (Gibson et al., 2015 [DOI](#)). This assay, coined ReVSA (repetitive visual stimulus-induced arousal) by the authors, had the advantage of scalability, while constraining flies to a walking arena that somewhat restricted the remarkably rich escape types flies otherwise exhibit. Here we carried this idea one step further by using a screen to present the shadows instead of a physically moving paddle and putting individual flies to linear corridors instead of the common circular fly arena. This ensured that the shadow reached the same coordinates in all linear tracks concurrently and made it easy to accurately determine when individual flies encountered the stimulus, aiding data analysis and scalability. We found the same escape behavioral repertoire as in studies with looming stimuli and ReVSA (Gibson et al., 2015 [DOI](#); Zacarias et al., 2018 [DOI](#)), with a similar dependence on walking speed (Zacarias et al., 2018 [DOI](#); Oram and Card, 2022 [DOI](#)), confirming that looming stimuli and passing shadows can both be considered as threat-inducing visual stimuli.

Mutations in the *PARK2* gene lead to impaired ubiquitination and a consequential mitochondrial dysfunction (Guo, 2010 [DOI](#), 2012 [DOI](#)), and early-onset PD in human patients. This led to the widespread use of Parkin flies as a genetic model of PD (Guo, 2010 [DOI](#); Chambers et al., 2013 [DOI](#);

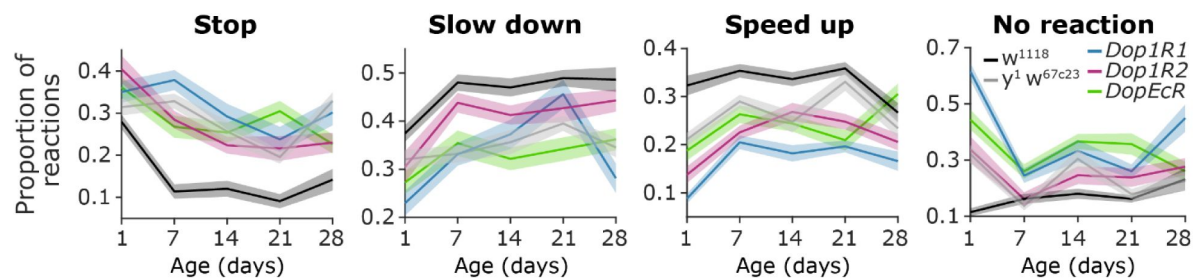


Fig. 5.

Changes in escape behavior from the first to the fourth week of life

Proportion of responses as a function of age for different groups of mutants and controls. From left to right, stop, slow down, speed up and no reaction trials are quantified. Lines and errorshades show mean and standard error. Exact genotypes: *iso* w^{1118} ; +; +; *Ddc-Gal4/+*; *Parkin*: +; +; *Ddc-Gal4/UAS-Parkin-R275W*; α -Syn: +; +; *Ddc-Gal4/UAS- α -Syn-A53T*. $y^1 w^{67c23}$; $y^1 w^{67c23}$. *Dop1R*: $y^1 w^*$; *Mi(MIC)Dop1R1^{M104437}*. *Dop1R2*: $y^1 w^*$; *Mi(MIC)Dop1R2^{M108664}*. *DopEcR*: w^{1118} ; *PBac{PB}DopEcR^{C02142}/TM6B*, *Tb¹*.

Aggarwal et al., 2019 [↗](#)). Of note, flies that lack Parkin display a significant degeneration of dopaminergic neurons (Whitworth et al., 2005 [↗](#)). We found similarities between Parkin and *Dop1R* mutant flies in their altered responses to predator-mimicking passing shadows: reduced walking speed and decreased overall reactivity, suggesting that the lack of dopamine action through *Dop1R* may be one of the common pathways underlying motor deficits. This is in line with a recent study demonstrating impaired startle-induced geotaxis and locomotor reactivity in *Dop1R* mutant flies (Sun et al., 2018 [↗](#)), similar to what had been shown for Parkin flies (Aggarwal et al., 2019 [↗](#)), and a demonstration of dopaminergic control over walking speed (Marquis and Wilson, 2022 [↗](#)). In contrast, flies expressing human α -syn showed moderate changes in motor behavior except for a marked prolongation of freezing duration upon threatening stimuli. This is in accordance with studies suggesting that α -syn misexpression is not fully penetrant under some conditions (Pesah et al., 2005 [↗](#); Nagoshi, 2018 [↗](#)), and calls for the use of other genetic *Drosophila* models, including Parkin flies, in studying the pathophysics of human PD.

The dopamine/ecdyseroid receptor *DopEcR* is a G-protein-coupled dual receptor for dopamine and the steroid hormone ecdysone. It has been proposed that the *DopEcR* may serve as an integrative hub for dopamine-mediated actions and stress responses in fruit flies (Petrucci et al., 2020 [↗](#)). We found that *DopEcR* mutant flies showed decreased mean walking speed, decreased probability of slowing down and decreased overall behavioral reactivity in response to predator-mimicking passing shadows. This pattern of alterations was largely similar to those observed in Parkin and *Dop1R* mutant flies, suggesting that the *DopEcR* may convey similar dopamine-mediated functions as *DopR1* at least in those motor aspects tested in the present study. A recent work demonstrated that serotonin also modulates both walking speed and startle response in flies, suggesting a complex neuromodulatory control over *Drosophila* motor behavior (Howard et al., 2019 [↗](#)). We recognize a limitation in using PBac{PB}DopEcRc02142 over the TM6B, Tb1 balancer chromosome, as the balancer itself may induce behavioral deficits in flies. We consider this unlikely, as the PBac{PB}DopEcRc02142 mutation demonstrates behavioral effects even in heterozygotes (Ishimoto et al., 2013 [↗](#)). Additionally, to our knowledge, no studies have reported behavioral deficits in flies carrying the TM6B, Tb1 balancer chromosome over a wild-type chromosome.

In humans, degeneration of midbrain dopaminergic neurons has been found as a direct cause of PD symptoms and DA supplementation with its precursor levodopa has been one of the most successful therapeutic approaches to date. The fruit fly and mammalian DA systems show a number of homologies which are thought to serve as a good basis for *Drosophila* PD models (Feany and Bender, 2000 [↗](#); Hewitt and Whitworth, 2017 [↗](#)). Specifically, a deep-running evolutionary conservation has been revealed between the arthropod central complex and the vertebrate basal ganglia, where GABAergic and dopaminergic neurons play key roles in motor control in both phylae (Strausfeld and Hirth, 2013 [↗](#)). Clusters of fly dopaminergic neurons that project to the ellipsoid body, the fan-shaped body and the lateral accessory lobes are thought to share homologies with the striatum-projecting dopaminergic neurons of the substantia nigra in mammals. Both the fruit fly central complex and vertebrate basal ganglia mediate a range of functions from motor control and sensorimotor integration to action selection and decision making to motivation (Strauss, 2002 [↗](#); Claridge-Chang et al., 2009 [↗](#); Kong et al., 2010 [↗](#)) and learning (McCurdy et al., 2021 [↗](#); Zolin et al., 2021 [↗](#); Fisher et al., 2022 [↗](#); Qiao et al., 2022 [↗](#); Taisz and Jefferis, 2022 [↗](#); Yamada et al., 2023 [↗](#)). It has been found that fruit flies lacking the DA-synthesizing tyrosine hydroxylase enzyme in the central nervous system show reduced activity and locomotor deficit that worsen with age, and also exhibit marked impairments in associative learning based on both appetitive and aversive reinforcement (Riemensperger et al., 2011 [↗](#)). This is in line with mammalian studies emphasizing the role of the dopaminergic nigrostriatal pathway in controlling goal-directed action (Eban-Rothschild et al., 2016 [↗](#)), reinforcement learning (Schultz et al., 1997 [↗](#); Lak et al., 2014 [↗](#)), and more recently, conveying aversive information (Menegas et al., 2015 [↗](#), 2018 [↗](#)). Our findings are consistent with the apparent homologies across fruit fly and mammalian dopaminergic systems and suggest that a more in-depth investigation of specific

behavioral changes and underlying dopamine-related dysfunctions may yield translatable results. We also observed age-related changes in motor behavior in response to threatening stimuli, particularly in 4-week-old flies, which may parallel the age-dependent worsening of PD symptoms in humans.

Methods

Animals

We used 2-weeks-old *Drosophila melanogaster* (both males and females) raised at 24°C and 60% humidity in a natural light-dark cycle for experiments presented in **Figures 1–4**. For age-group comparisons (**Fig. 5**), fruit flies aged 1, 7, 14, 21 and 28 days were used. These flies were raised at 29°C and 70% humidity to accelerate aging.

Three sets of mutant groups were used for behavioral comparison. In the first set of animals we used w^* ; *UAS-Parkin-R275W* (created in the laboratory of Kah-Leong Lim, Neurodegeneration Research Laboratory, National Neuroscience Institute, Singapore) (Wang et al., 2007; Kovács et al., 2017) and w^* ; *UAS-alpha-Syn-A53T* (BDSC 8148) from the Eotvos Lorand University (Szinyákóvics et al., 2023), with y^1 , v^1 ; *UAS-GFP* (BDSC 35786) and $+$; *P{Ddc-GAL4.L}4.36* (modified version of BDSC 7009, the original first chromosome w^{1118} has been replaced to w^+ in this study) as controls. For the second set of animals we used *Dop1R1*, *Dop1R2* and *DopEcR* mutant flies kindly donated by the Anne Von Phillipsborn lab, DANDRITE, Aarhus University and y^1 w^* ; *Mi{MIC}Dop1R1^{M104437}* (BDSC 43773), y^1 w^* ; *Mi{MIC}Dop1R2^{M108664}* (BDSC 51098), and w^{1118} ; *PBac{PB}DopEcR^{C02142/TM6B}*, *Tb1* (BDSC 10847) with w^{1118} (BDSC 5905) y^1 and y^1 w^{67c23} (BDSC 6599) as controls from the Bloomington stock center. **Table 1** shows the number of recorded flies for the age group comparisons, and **Table 2** shows the number of recorded flies for comparing the mutant groups. For the third set of flies we used $w[*]$; *P{w[+mC]=ple-GAL4.F}3* (*TH-Gal4*), Bloomington Drosophila Stock Center: 8848 (Friggi-Grelín et al., 2003), and $y[*]$ $w[*]$; *P{w[+mW.hs]=GawB}NP6510* (*NP6510-Gal4*), Kyoto DGGR: 113956 (Riemensperger et al., 2013; Majcin Dorcikova et al., 2023). Since it was important to ensure that the animals' light perception was not impaired, we replaced the white loss-of-function mutant allele of the transgenic *Gal4* lines with a wild-type allele. As controls, we used transheterozygous $+/TH-Gal4$ and $+/NP6510-Gal4$ strains in all cases.

Arena

We designed an arena made of four layers of plexiglass and metal that formed 13 tunnels (**Fig. 1a**). Each tunnel was 52mm x 5mm x 1.5mm, in which individual flies were able to move freely in two dimensions but could not fly. The bottom layer was a standard transparent plexiglass layer. The second layer contained the tunnel walls made of metal to prevent horizontal spread of light. The third layer was the top of the tunnels, made of transparent plexiglass, which could slide above the tunnel. It also contained a small hole on the side, where the flies could be inserted into the tunnel through a pipette. The fourth layer was a metal cover designed with cut-outs that matched the shape and position of the tunnels to prevent flies in the neighboring tunnels from detecting the shadow before it reached their position.

Movement tracking

Flies were tracked by using a FLIR Camera (FLIR Blackfly S USB3 FLIR Systems, Wilsonville, OR, US) placed under the arena (**Fig. 1a**). The frame rate was set to 100 frames/second to allow the detection of position with high temporal resolution. The recorded images were processed by Bonsai (2.5.2) tracking software (Lopes et al., 2015), using customized Bonsai code. The code extracts each tunnel area separately from the image and detects flies by selecting the biggest and

Table 1.

Number of recorded flies for age group comparisons.

	<i>Dop1R1</i>	<i>Dop1R2</i>	<i>DopEcR</i>	<i>w</i> ¹¹¹⁸	<i>y</i> ¹ <i>w</i> ^{67c23}
1-day-old	35	36	36	35	41
1-week-old	36	54	36	36	36
2-week-old	36	36	36	36	30
3-week-old	35	35	29	36	36
4-week-old	30	36	34	30	30

Table 2.

Number of recorded flies for comparing the mutant groups.

2-week-old	No. of flies
<i>Dop1R1</i>	45
<i>Dop1R2</i>	47
<i>DopEcR</i>	35
<i>w</i> ¹¹¹⁸	46
<i>y</i> ¹ , <i>w</i> ^{67c23}	40
Parkin (Ddc)	78
α-Syn (Ddc)	69
<i>iso w</i> ¹¹¹⁸	92
+GFP	65
Parkin (NP6510)	170
α-Syn (NP6510) ₋	181
+/ <i>NP6510-Gal4</i>	176
Parkin (TH)	322
α-Syn (TH)	176
+/ <i>TH-Gal4</i>	282

darkest pixel object in the tunnel area in real time. Fly position coordinates were calculated from the centroid. The software stored x-y position coordinates along with their timestamps in csv files, which were later processed in Matlab R2016a (Mathworks, Natick, MA, US).

Predator-mimicking shadow stimuli

A 10.1-inch screen (HannStar HSD101PWW1-A00,) was placed on the top of the arena. The screen presented a yellow background. We implemented ‘shadow’ stimuli in red color, as these stimuli were suggested to be perceived by fruit flies as dark shadows and also enabled continuous tracking of the animals (Sharkey et al., 2020 [↗](#)). The red color screen represented a large enough contrast change to evoke escape behaviors of flies but also provided sufficient brightness to enable continuous motion tracking. A passing shadow stimulus was chosen, as it was affecting all tunnels simultaneously. To mimic an approaching shadow, a red screen slid in from the side, stayed on for 2 seconds, then slid out. The shadow stimuli were separated by pseudorandomized inter-trial intervals with an approximately exponential distribution with a mean of 52.2 seconds, preventing flies from anticipating the next shadow stimulus.

Identification of behavioral response types to the threatening stimuli

Animal speed was calculated as a function of time in a -200 ms to 1000 ms window around each shadow stimulus in 10 ms time bins. These speed functions were normalized by subtracting the average speed in the 200 ms window before the shadow presentation to reveal stimulus induced absolute instantaneous speed changes. Principal component analysis (PCA) was used to reduce dimensionality of the normalized speed functions in the 1 s interval following shadow presentation in a way that the variance between trials is maximally preserved in the low-dimensional representation. Agglomerative hierarchical clustering was performed in the space spanned by the first three principal components to identify trials with distinct characteristic response types.

Behavior detection

After identifying the four most frequently observed animal responses to shadow presentations, we algorithmically defined them based on the speed and acceleration thresholds. First, we examined the speed of the animal in the 200 ms time window before the shadow (v_{pre}); stopping/freezing or slowing down was only possible if the fly had been moving, defined by $v_{pre} > 0$ mm/s. Second, we examined the acceleration (a_{post}) in the 1 s interval following the shadow using a 100 ms moving average window to characterize the flies’ response. Reactions were considered robust if the absolute value of the acceleration reached 200 mm/s². A trial was characterized as a ‘stop’ trial, if the average v_{pre} was above 0 mm/s, the fly’s first reaction was deceleration, and 0 mm/s speed was reached before accelerating again. If the first reaction was deceleration but 0 mm/s speed was not reached, the trial was classified as a ‘slow down’ trial. In ‘speed up’ trials, the first response following the shadow was acceleration. If the absolute value of a_{post} did not reach the 200 mm/s² threshold during the 1 s interval following the shadow, the trial was labeled as a ‘no reaction’ trial.

Data analysis and statistics

Data analysis was performed in Matlab R2016a (Mathworks, Natick, MA, US) using custom-written code. Statistical comparison between groups of flies was performed by two-sided Mann-Whitney U-test. Exact p values are reported in the Results section. We performed multi-group comparison in the ‘Changes in escape behavior from the first to the fourth week of life’ section using two-way ANOVA, as it is reportedly robust to non-normality of the data (Mooi et al., 2018 [↗](#); Knief and Forstmeier, 2021 [↗](#)), while there is a lack of a consensual non-parametric alternative. Nevertheless, Kruskal-Wallis tests for the main effect of age showed significant effects in all genotypes in accordance with the ANOVA, confirming the results (Stop frequency, *DopEcR* p =

0.0007; *Dop1R1*, $p = 0.004$; *Dop1R2*, $p = 9.94 \times 10^{-5}$; w^{1118} , $p = 9.89 \times 10^{-13}$; $y^1 w^{67c23}$, $p = 2.54 \times 10^{-5}$; Slowing down frequency, *DopEcR*, $p = 0.0421$; *Dop1R1*, $p = 5.77 \times 10^{-6}$; *Dop1R2*, $p = 0.011$; w^{1118} , $p = 2.62 \times 10^{-5}$; $y^1 w^{67c23}$, $p = 0.0382$; Speeding up frequency, *DopEcR*, $p = 0.0003$; *Dop1R1*, $p = 2.06 \times 10^{-7}$; *Dop1R2*, $p = 2.19 \times 10^{-6}$; w^{1118} , $p = 0.0044$; $y^1 w^{67c23}$, $p = 1.36 \times 10^{-5}$). We used Mann-Whitney U-test for post hoc analyses to maintain consistent use of tests across figures. Since there is no straightforward way of correcting for multiple comparisons in this case, we reported uncorrected p values and suggest considering them at $p = 0.01$, which minimizes type I errors. Nevertheless, Tukey's tests with the 'honest significance' approach yielded similar results. The effect of movement speed on the distribution of behavioral response types was tested using a nested Monte Carlo simulation framework (**Fig. S5** [↗](#)). This simulation aimed to model how different movement speeds impact the probability distribution of response types, comparing these simulated outcomes to empirical data. This approach allowed us to determine whether observed differences in response distributions were solely due to speed variations across genotypes or if additional behavioral factors contributed to the differences. First, we calculated the probability of each response type at different specific speed values (outer model). These probabilities were derived from the grand average of all trials across each genotype, capturing the overall tendency at various speeds. Second, we simulated behavior of virtual flies ($n = 3000$ per genotypes, which falls within the same order of magnitude as the number of experimentally recorded trials from different genotypes) by drawing random velocity values from the empirical velocity distribution specific to the given genotype and then randomly selecting a reaction based on the reaction probabilities associated with the drawn velocity (inner model). Finally, we calculated reaction probabilities for the virtual flies and compared them with real data from animals of the same genotype. Differences were statistically tested by Chi-squared test.

Supplementary Figure Legends

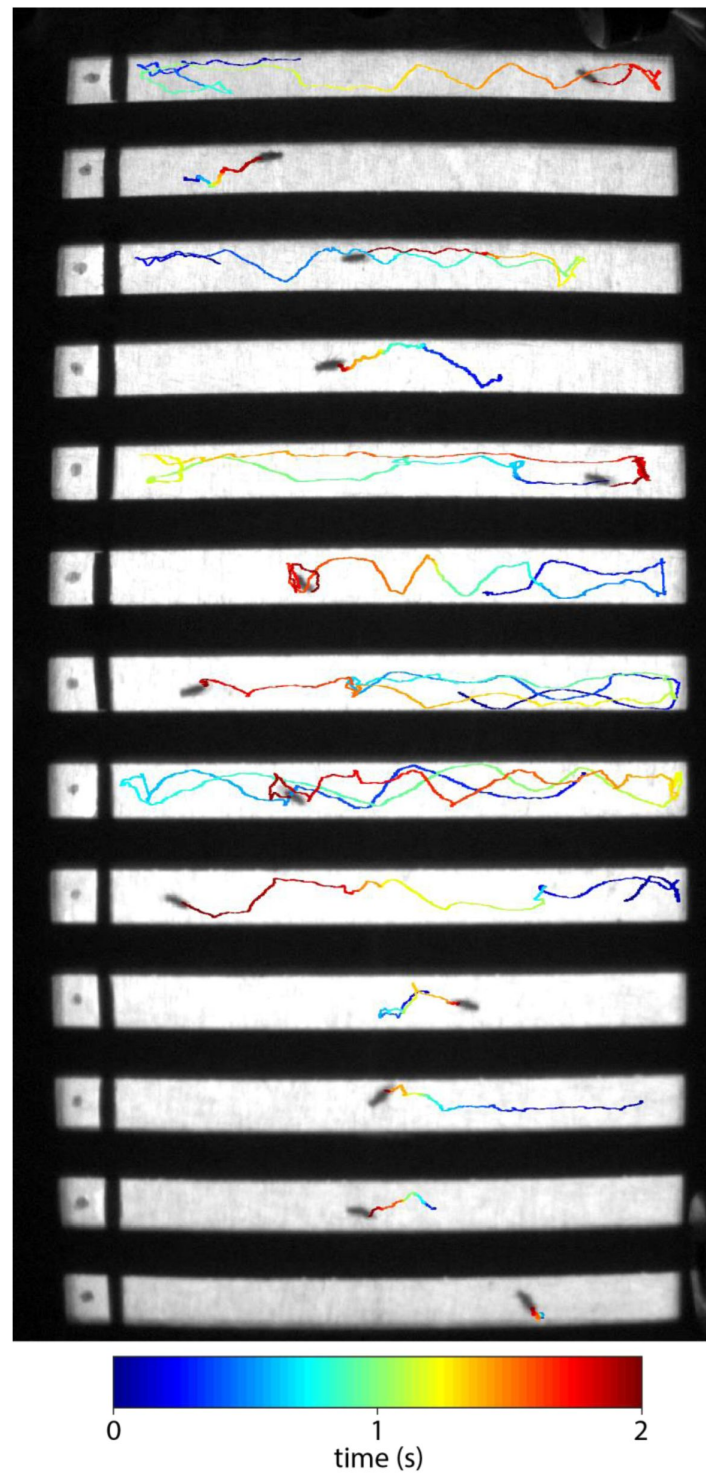


Fig. S1.

Picture of the arena from the camera view

The 13 tunnels of the arena from the camera view with the tracking lines showing that the animals can move in 2 dimensions. The tracking line color indicates time, with dark blue and red marking the beginning and the end of the tracking, respectively.

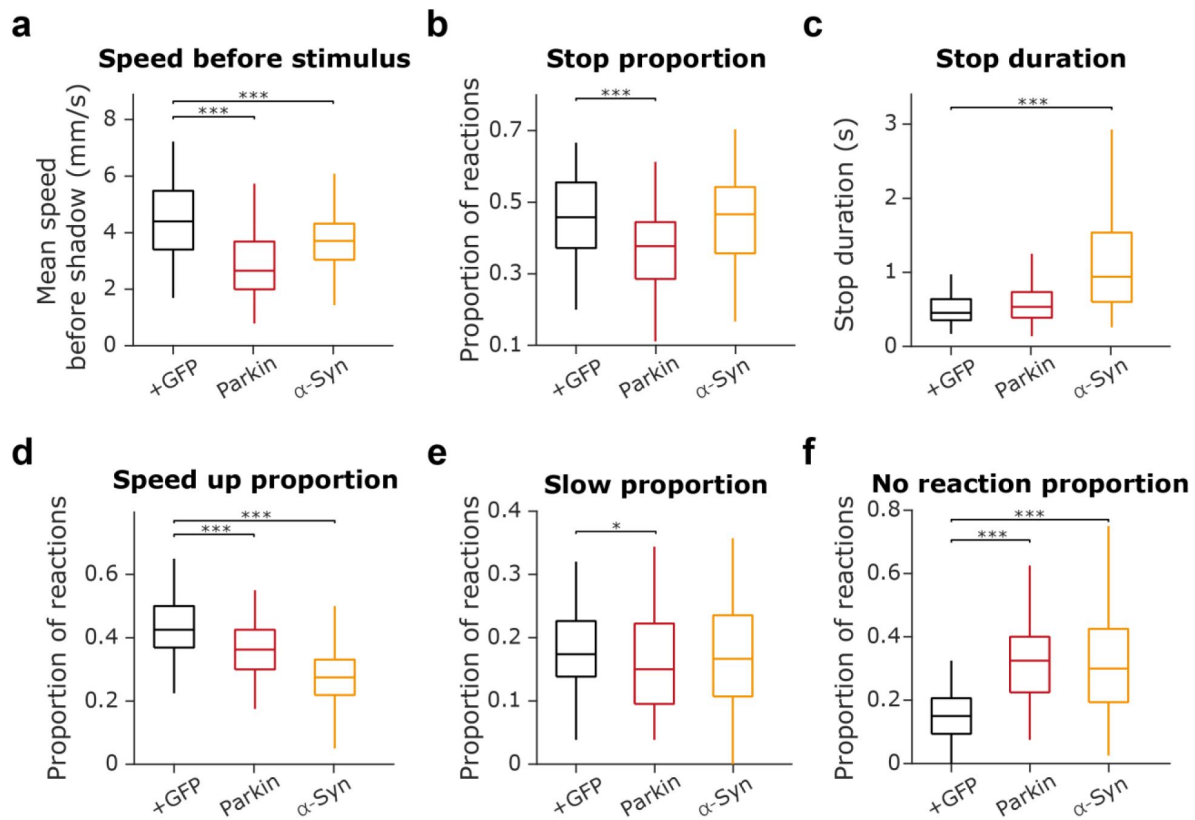


Fig. S2.

Parkin and α -Syn fly group escape reaction compared to GFP mutant flies

a. Distribution of the mean speed measured in the 0.2 second before shadow presentation for Parkin, α -Syn and +GFP mutant groups. Parkin and α -Syn flies showed reduced mean speed compared to +GFP controls (Parkin vs. +GFP, $p = 1.58 \times 10^{-11}$; α -Syn vs. +GFP, $p = 2.85 \times 10^{-4}$; Mann-Whitney U-test). **b.** Distribution of the proportion of stop trials in different mutant groups. Parkin flies showed a reduced tendency for stopping compared to the +GFP group (Parkin vs. +GFP, $p = 7.1 \times 10^{-5}$; α -Syn vs. +GFP, $p = 0.67$; Mann-Whitney U-test). **c.** Distribution of stop duration in different mutant groups. α -Syn flies showed increased stop durations compared to GFP (Parkin vs. +GFP, $p = 0.17$; α -Syn vs. +GFP, $p = 3.98 \times 10^{-10}$; Mann-Whitney U-test). **d.** Distribution of the proportion of speed up trials in different mutant groups. Both Parkin and α -Syn flies showed a reduced tendency to speed up compared to +GFP controls (Parkin vs. +GFP, $p = 8.3 \times 10^{-5}$; α -Syn vs. +GFP, $p = 1.96 \times 10^{-12}$; Mann-Whitney U-test). **e.** Distribution of the proportion of slow down trials in different mutant groups. Parkin flies showed a reduced tendency to slow down compared to +GFP controls (Parkin vs. +GFP, $p = 0.044$; α -Syn vs. +GFP, $p = 0.465$; Mann-Whitney U-test). **f.** Distribution of the proportion of 'no reaction' trials in different mutant groups. Both Parkin and α -Syn flies showed reduced reactivity compared to +GFP controls (Parkin vs. +GFP, $p = 9.4 \times 10^{-13}$; α -Syn vs. +GFP, $p = 1.3 \times 10^{-8}$; Mann-Whitney U-test). *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. Exact genotypes: *iso w*¹¹¹⁸: +; +; *Ddc-Gal4*/+. Parkin: +; +; *Ddc-Gal4/UAS-Parkin-R275W*. α -Syn: +; +; *Ddc-Gal4/UAS- α -Syn-A53T*.

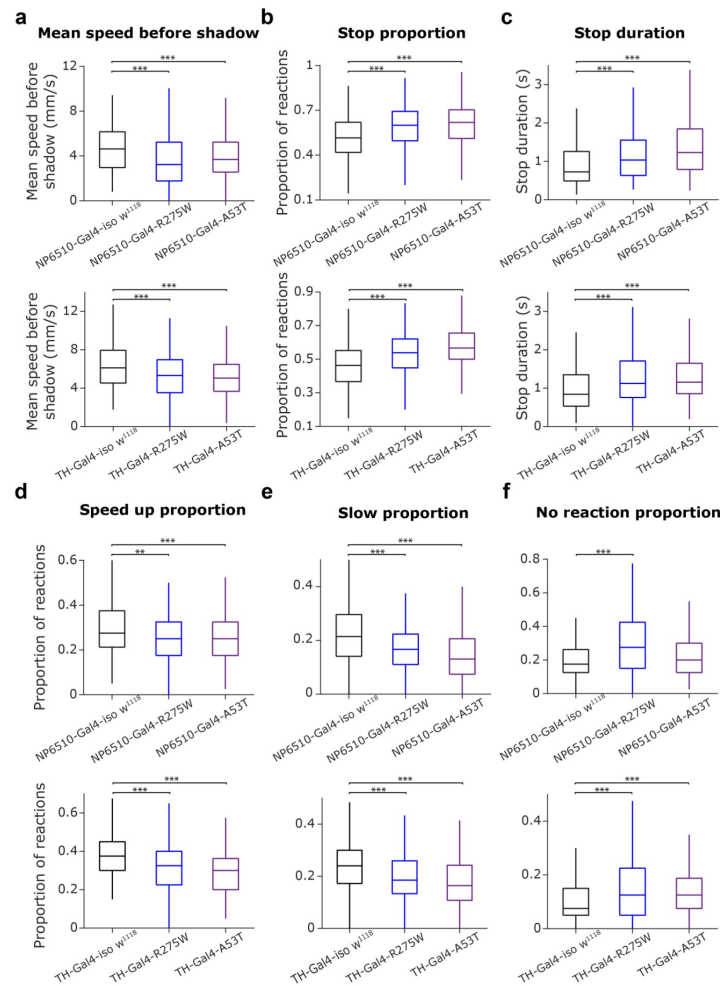


Fig. S3.

Restricting Parkin and α -Syn mutations to dopaminergic neurons

a, Distribution of the mean speed measured in the 0.2 second before shadow presentation for Parkin (R275W), α -Syn (A53T) and control (*iso w¹¹¹⁸*) groups. Top, *NP6510-Gal4* lines; bottom, *TH-Gal4* lines in all panels. *NP6510-Gal4-R275W*, 30.3% decrease compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 1.55 \times 10^{-6}$; *NP6510-Gal4-A53T*, 20.4% decrease compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 5.82 \times 10^{-10}$; *TH-Gal4-R275W*, 13.08% decrease compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 1.49 \times 10^{-6}$; *TH-Gal4-A53T*, 17.57% decrease compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 6.75 \times 10^{-7}$; Mann-Whitney U-test. **b**, Distribution of the proportion of stop trials in different mutant groups. *NP6510-Gal4-R275W*, 16.47% increase compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 4.4 \times 10^{-11}$; *NP6510-Gal4-A53T*, 20.17% increase compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 1.38 \times 10^{-17}$; *TH-Gal4-R275W*, 16.32% increase compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 1.55 \times 10^{-4}$; *TH-Gal4-A53T*, 22.41% increase compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 8.82 \times 10^{-10}$; Mann-Whitney U-test. **c**, Distribution of stop duration in different mutant groups. *NP6510-Gal4-R275W*, 42.6% increase compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 5.99 \times 10^{-5}$; *NP6510-Gal4-A53T*, 69.5% increase compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 4.33 \times 10^{-11}$; *TH-Gal4-R275W*, 33.62% increase compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 1.63 \times 10^{-6}$; *TH-Gal4-A53T*, 37.48% increase compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 3.89 \times 10^{-7}$; Mann-Whitney U-test. **d**, Distribution of the proportion of speed up trials in different mutant groups. *NP6510-Gal4-R275W*, 9.1% decrease compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 0.0052$; *NP6510-Gal4-A53T*, 9.1% decrease compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 3.14 \times 10^{-4}$; *TH-Gal4-R275W*, 13.3% decrease compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 1.25 \times 10^{-9}$; *TH-Gal4-A53T*, 20% decrease compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 1.33 \times 10^{-15}$; Mann-Whitney U-test. **e**, Distribution of the proportion of slow down trials in different mutant groups. *NP6510-Gal4-R275W*, 22.2% decrease compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 8.23 \times 10^{-5}$; *NP6510-Gal4-A53T*, 39.1% decrease compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 2.42 \times 10^{-10}$; *TH-Gal4-R275W*, 22.8% decrease compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 2.85 \times 10^{-8}$; *TH-Gal4-A53T*, 31.49% decrease compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 1.95 \times 10^{-11}$; Mann-Whitney U-test. **f**, Distribution of the proportion of 'no reaction' trials in different mutant groups. *NP6510-Gal4-R275W*, 57.14% increase compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 3.49 \times 10^{-8}$; *NP6510-Gal4-A53T*, 14.28% increase compared to *NP6510-Gal4-iso w¹¹¹⁸*, $p = 0.1658$; *TH-Gal4-R275W*, 66.7% increase compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 2.77 \times 10^{-10}$; *TH-Gal4-A53T*, 66.7% increase compared to *TH-Gal4-iso w¹¹¹⁸*, $p = 1.11 \times 10^{-6}$; Mann-Whitney U-test.

Fig. S4.

Stop duration showed large variance in α -Syn flies

a, Distributions of stop duration in *iso w¹¹¹⁸* controls, Parkin and α -Syn flies. **b**, Permutation tests revealed a significantly increased variance in α -Syn compared to control flies ($p = 0.0001$). Histograms show shuffled variance difference between Parkin (left) or α -Syn (right) and control groups. Observed difference in variance is indicated by a blue vertical line. The critical value corresponding to $p = 0.05$ is indicated by a green dashed vertical line.

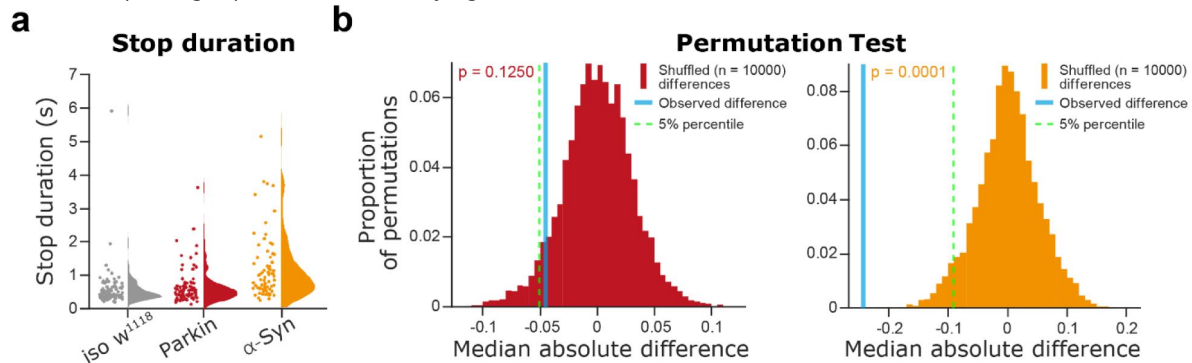
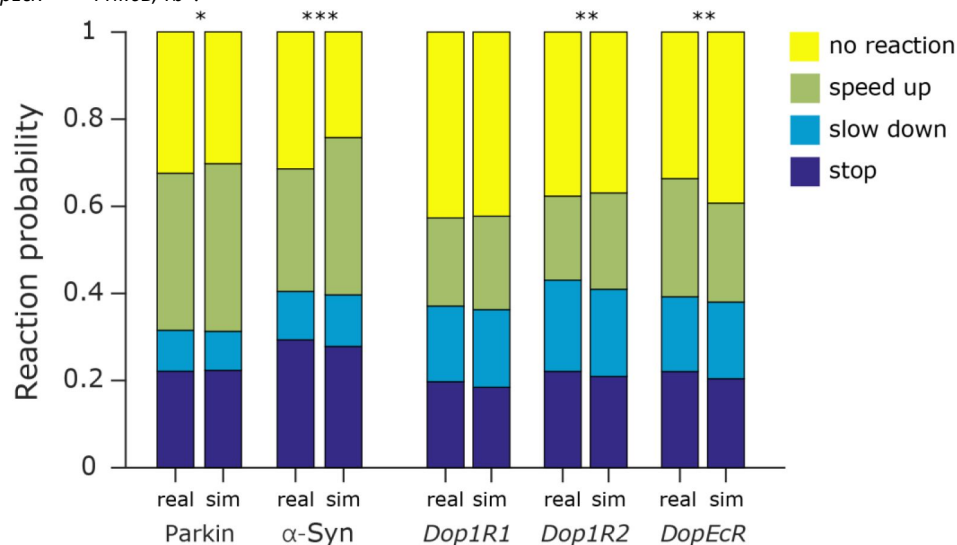


Fig. S5.

Differences in walking speed do not explain different escape behaviors

Real and simulated proportion of response types are shown for the mutant groups tested. For the simulations, random velocity values were drawn from the velocity distribution of each genotype and then a response type was randomly selected based on the response type distributions associated with the drawn velocity. Bar plots show the proportion of response types color-coded. For each mutant group, the right bar shows the measured proportion of escape responses, while the left bar shows the simulated distribution. We found significant differences between the real and simulated distributions for the following groups: Parkin ($p = 0.0487$), α -Syn ($p < 0.000001$), *Dop1R2* ($p = 0.006157$) and *DopEcR* ($p = 0.003901$). We found no significant differences for the *Dop1R1* group ($p = 0.332$; Chi-square test for all the statistics). *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. Exact genotypes: *iso w¹¹¹⁸*: +; +; *Ddc-Gal4*/+. Parkin: +; +; *Ddc-Gal4/UAS-Parkin-R275W*. α -Syn: +; +; *Ddc-Gal4/UAS- α -Syn-A53T*. *y¹ w^{67c23}*: *y¹ w^{67c23}*. *Dop1R*: *y¹ w**; *MI(MIC)Dop1R1^{MI04437}*. *Dop1R2*: *y¹ w**; *MI(MIC)Dop1R2^{MI08664}*. *DopEcR*: *w¹¹¹⁸*; *PBac{PB}DopEcR^{c02142}/TM6B, Tb¹*.



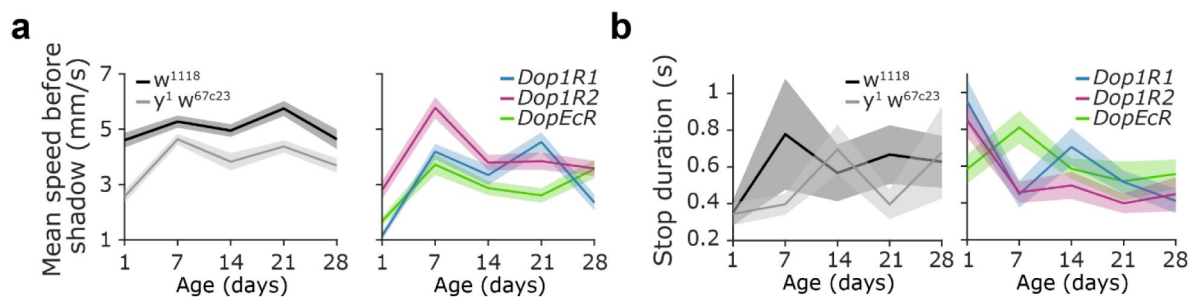


Fig. S6.

Changes in escape behavior from the first to the fourth week of life

a, Average walking speed as a function of age for different groups of controls (left) and mutants (right). Lines and errorshades show mean and standard error. Mean speed showed significant differences among age groups (two-way ANOVA, age, $f = 41.38$, $p = 9.39 \times 10^{-32}$; genotype, $f = 42.19$, $p = 2.46 \times 10^{-32}$; genotype $\times$ age, $f = 4.36$, $p = 2.667 \times 10^{-8}$; Tukey's post hoc test: 1 day old vs. 1 week old, $p = 9.92 \times 10^{-9}$; 1 week old vs. 2 weeks old, $p = 7.34 \times 10^{-7}$; 2 weeks old vs. 3 weeks old, $p = 0.085$; 3 weeks old vs. 4 weeks old, $p = 0.0043$). **b**, Stop duration as a function of age for different groups of controls (left) and mutants (right). We found that age did not effect the duration of stops (two-way ANOVA, age, $f = 0.85$, $p = 0.49$; genotype, $f = 0.91$, $p = 0.46$; genotype $\times$ age, $f = 2.53$, $p = 0.0008$; Tukey's post hoc test: 1 day old vs. 1 week old, $p = 0.98$; 1 week old vs. 2 weeks old, $p = 0.992$; 2 weeks old vs. 3 weeks old, $p = 0.578$; 3 weeks old vs. 4 weeks old, $p = 0.9759$). Lines and errorshades show mean and standard error.

Data and code availability

All analysis functions are available at <https://github.com/hangyabalazs/Drosophila-behavior-analysis>. All data are available at <https://figshare.com/s/40539ee040a269bf05e5>.

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Additional files

Supplementary Movie 2. Example recording of Parkin flies in the arena. Similar behaviors can be observed as in *iso w¹¹¹⁸* flies (see Supplementary Movie 1). [↗](#)

Supplementary Movie 1. Example recording of *iso w¹¹¹⁸* flies in the arena. Flies could walk freely inside the tunnels, speed up, slow down, stop, turn, climb the side walls, flip, or groom their wings. [↗](#)

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Reviewer #1 (Public review):

Translating discoveries from model organisms to humans is often challenging, especially in neuropsychiatric diseases, due to the vast gaps in the circuit complexities and cognitive capabilities. Kajtor et al. propose to bridge this gap in the fly models of Parkinson's disease (PD) by developing a new behavioural assay where flies respond to a moving shadow by modifying their locomotor activities. The authors believe the flies' response to the shadow approximates their escape response to an approaching predator. To validate this argument, they tested several PD-relevant transgenic fly lines and showed that some of them indeed have altered responses in their assay.

Strengths:

This single-fly-based assay is easy and inexpensive to set up, scalable and provides sensitive, quantitative estimates to probe flies' optomotor acuity. The behavioural data is detailed, and the analysis parameters are well-explained.

Weaknesses:

The authors have yet to link cellular physiology to behaviour. It will be interesting to see how future use of this assay helps uncover connections between cellular pathology and behavioural changes.

<https://doi.org/10.7554/eLife.90905.2.sa2>

Reviewer #2 (Public review):

The manifestation and progression of neurodegenerative disorders is poorly understood. Many of the neuronal disorders start by presenting subtle changes in neuronal circuit and quantification and measurement of these subtle behavior responses could help one delineate the mechanisms involved. The present study very nicely uses the flies' behavioral response to predator-mimicking passing shadows to measure subtle changes in their behavior. The data from various fly genetic models of Parkinson's disease supports their claim. This single trial

method is useful to capture the individual animal's response to the threatening stimuli but stops short of capturing the fine ambulatory responses which could provide further information on an individual's behavioral response. By capturing the fine features, the authors could get detailed observations, such as posture, gait or wing positioning for a better understanding the behavioral response to the passing shadow.

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Author response:

The following is the authors' response to the original reviews

We thank the Reviewers for their constructive comments and the Editor for the possibility to address the Reviewers' points in this rebuttal. We

- (1) Conducted new experiments with *NP6510-Gal4* and *TH-Gal4* lines to address potential behavioral differences due to targeting dopaminergic vs. both dopaminergic and serotonergic neurons
- (2) Conducted novel data analyses to emphasize the strength of sampling distributions of behavioral parameters across trials and individual flies
- (3) Provided Supplementary Movies
- (4) Calculated additional statistics
- (5) Edited and added text to address all points of the Reviewers.

Please see our point-by-point responses below.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

Translating discoveries from model organisms to humans is often challenging, especially in neuropsychiatric diseases, due to the vast gaps in the circuit complexities and cognitive capabilities. Kajtor et al. propose to bridge this gap in the fly models of Parkinson's disease (PD) by developing a new behavioral assay where flies respond to a moving shadow by modifying their locomotor activities. The authors believe the flies' response to the shadow approximates their escape response to an approaching predator. To validate this argument, they tested several PD-relevant transgenic fly lines and showed that some of them indeed have altered responses in their assay.

Strengths:

This single-fly-based assay is easy and inexpensive to set up, scalable, and provides sensitive, quantitative estimates to probe flies' optomotor acuity. The behavioral data is detailed, and the analysis parameters are well-explained.

We thank the Reviewer for the positive assessment of our study.

Weaknesses:

While the abstract promises to give us an assay to accelerate fly-to-human translation, the authors need to provide evidence to show that this is indeed the case. They have used

PD lines extensively characterized by other groups, often with cheaper and easier-to-setup assays like negative geotaxis, and do not offer any new insights into them. The conceptual leap from a low-level behavioral phenotype, e.g. changes in walking speed, to recapitulating human PD progression is enormous, and the paper does not make any attempt to bridge it. It needs to be clarified how this assay provides a new understanding of the fly PD models, as the authors do not explore the cellular/circuit basis of the phenotypes. Similarly, they have assumed that the behavior they are looking at is an escape-from-predator response modulated by the central complex- is there any evidence to support these assumptions? Because of their rather superficial approach, the paper does not go beyond providing us with a collection of interesting but preliminary observations.

We thank the Reviewer for pointing out some limitations of our study. We would like to emphasize that what we perceive as the main advantage of performing single-fly and single-trial analyses is the access to rich data distributions that provide more fine-scale information compared to bulk assays. We think that this is exactly going one step closer to ‘bridging the enormous conceptual leap from a low-level behavioral phenotype, e.g. changes in walking speed, to recapitulating human PD progression’, and we showcase this in our study by comparing the distributions over the entire repertoire of behavioral responses across fly mutants. Nevertheless, we agree with the Reviewer that many more steps in this direction are needed to improve translatability. Therefore, we toned down the corresponding statements in the Abstract and in the Introduction. Moreover, to further emphasize the strength of sampling distributions of behavioral parameters across trials and individual flies, we complemented our comparisons of central tendencies with testing for potential differences in data dispersion, demonstrated in the novel Supplementary Figure S4.

Looming stimuli have been used to characterize flies’ escape behaviors. These studies uncovered a surprisingly rich behavioral repertoire (Zacarias et al., 2018), which was modulated by both sensory and motor context, e.g. walking speed at time of stimulus presentation (Card and Dickinson, 2008; Oram and Card, 2022; Zacarias et al., 2018). The neural basis of these behaviors was also investigated, revealing loom-sensitive neurons in the optic lobe and the giant fiber escape pathway (Ache et al., 2019; de Vries and Clandinin, 2012). Although less frequently, passing shadows were also employed as threat-inducing stimuli in flies (Gibson et al., 2015). We opted for this variant of the stimulus so that we could ensure that the shadow reached the same coordinates in all linear track concurrently, aiding data analysis and scalability. Similar to the cited study, we found the same behavioral repertoire as in studies with looming stimuli, with an equivalent dependence on walking speed, confirming that looming stimuli and passing shadows can both be considered as threat-inducing visual stimuli. We added a discussion on this topic to the main text.

Reviewer #2 (Public Review):

*In this study, Kajtor et al investigated the use of a single-animal trial-based behavioral assay for the assessment of subtle changes in the locomotor behavior of different genetic models of Parkinson's disease of *Drosophila*. Different genotypes used in this study were *Ddc-GAL4>UASParkin-275W* and *UAS- α -Syn-A53T*. The authors measured *Drosophila*'s response to predator-mimicking passing shadow as a threatening stimulus. Along with these, various dopamine (DA) receptor mutants, *Dop1R1*, *Dop1R2* and *DopEcR* were also tested.*

The behavior was measured in a custom-designed apparatus that allows simultaneous testing of 13 individual flies in a plexiglass arena. The inter-trial intervals were randomized for 40 trials within 40 minutes duration and fly responses were defined into freezing, slowing down, and running by hierarchical clustering. Most of the mutant flies

showed decreased reactivity to threatening stimuli, but the speed-response behavior was genotype invariant.

These data nicely show that measuring responses to the predator-mimicking passing shadows could be used to assess the subtle differences in the locomotion parameters in various genetic models of Drosophila.

The understanding of the manifestation of various neuronal disorders is a topic of active research. Many of the neuronal disorders start by presenting subtle changes in neuronal circuits and quantification and measurement of these subtle behavior responses could help one delineate the mechanisms involved. The data from the present study nicely uses the behavioral response to predator-mimicking passing shadows to measure subtle changes in behavior. However, there are a few important points that would help establish the robustness of this study.

We thank the Reviewer for the constructive comments and the positive assessment of our study.

(1) The visual threat stimulus for measuring response behavior in Drosophila is previously established for both single and multiple flies in an arena. A comparative analysis of data and the pros and cons of the previously established techniques (for example, Gibson et al., 2015) with the technique presented in this study would be important to establish the current assay as an important advancement.

We thank the Reviewer for this suggestion. We included the following discussion on measuring response behavior to visual threat stimuli in the revised manuscript.

Many earlier studies used looming stimulus, that is, a concentrically expanding shadow, mimicking the approach of a predator from above, to study escape responses in flies (Ache et al., 2019; Card and Dickinson, 2008; de Vries and Clandinin, 2012; Oram and Card, 2022; Zacarias et al., 2018) as well as rodents (Braine and Georges, 2023; Heinemans and Moita, 2024; Lecca et al., 2017). These assays have the advantage of closely resembling naturalistic, ecologically relevant threatinducing stimuli, and allow a relatively complete characterization of the fly escape behavior repertoire. As a flip side of their large degree of freedom, they do not lend themselves easily to provide a fully standardized, scalable behavioral assay. Therefore, Gibson et al. suggested a novel threat-inducing assay operating with moving overhead translational stimuli, that is, passing shadows, and demonstrated that they induce escape behaviors in flies akin to looming discs (Gibson et al., 2015). This assay, coined ReVSA (repetitive visual stimulus-induced arousal) by the authors, had the advantage of scalability, while constraining flies to a walking arena that somewhat restricted the remarkably rich escape types flies otherwise exhibit. Here we carried this idea one step further by using a screen to present the shadows instead of a physically moving paddle and putting individual flies to linear corridors instead of the common circular fly arena. This ensured that the shadow reached the same coordinates in all linear tracks concurrently and made it easy to accurately determine when individual flies encountered the stimulus, aiding data analysis and scalability. We found the same escape behavioral repertoire as in studies with looming stimuli and ReVSA (Gibson et al., 2015; Zacarias et al., 2018), with a similar dependence on walking speed (Oram and Card, 2022; Zacarias et al., 2018), confirming that looming stimuli and passing shadows can both be considered as threat-inducing visual stimuli.

(2) Parkinson's disease mutants should be validated with other GAL-4 drivers along with DdcGAL4, such as NP6510-Gal4 (Riemensperger et al., 2013). This would be important to delineate the behavioral differences due to dopaminergic neurons and serotonergic neurons and establish the Parkinson's disease phenotype robustly.

We thank the Reviewer for point out this limitation. To address this, we repeated our key experiments in Fig.3. with both TH-Gal4 and NP6510-Gal4 lines, and their respective controls. These yielded largely similar results to the Ddc-Gal4 lines reported in Fig.3., reproducing the decreased speed and decreased overall reactivity of PD-model flies. Nevertheless, TH-Gal4 and NP6510-Gal4 mutants showed an increased propensity to stop. Stop duration showed a significant increase not only in α -Syn but also in Parkin fruit flies. These novel results have been added to the text and are demonstrated in Supplementary Figure S3.

(3) The DopEcR mutant genotype used for behavior analysis is w1118; PBac{PB}DopEcRc02142TM6B, Tb1. Balancer chromosomes, such as TM6B, Tb can have undesirable and uncharacterised behavioral effects. This could be addressed by removing the balancer and testing the DopEcR mutant in homozygous (if viable) or heterozygous conditions.

We appreciate the Reviewer's comment and acknowledge the potential for the *DopEcR* balancer chromosome to produce unintended behavioral effects. However, given that this mutant was not essential to our main conclusions, we opted not to repeat the experiment. Nevertheless, we now discuss the possible confounds associated with using the PBac{PB}DopEcRc02142 mutant allele over the balancer chromosome. "We recognize a limitation in using PBac{PB}DopEcRc02142 over the TM6B, *Tb*¹ balancer chromosome, as the balancer itself may induce behavioral deficits in flies. We consider this unlikely, as the PBac{PB}DopEcRc02142 mutation demonstrates behavioral effects even in heterozygotes (Ishimoto et al., 2013). Additionally, to our knowledge, no studies have reported behavioral deficits in flies carrying the TM6B, *Tb*¹ balancer chromosome over a wild-type chromosome."

(4) The height of the arena is restricted to 1mm. However, for the wild-type flies (Canton-S) and many other mutants, the height is usually more than 1mm. Also, a 1 mm height could restrict the fly movement. For example, it might not allow the flies to flip upside down in the arena easily. This could introduce some unwanted behavioral changes. A simple experiment with an arena of height at least 2.5mm could be used to verify the effect of 1mm height.

We thank the Reviewer for this comment, which prompted us to reassess the dimensions of the apparatus. The height of the arena was 1.5 mm, which we corrected now in the text. We observed that the arena did not restrict the flies walking and that flies could flip in the arena. We now include two Supplementary Movies to demonstrate this.

(5) The detailed model for Monte Carlo simulation for speed-response simulation is not described. The simulation model and its hyperparameters need to be described in more depth and with proper justification.

We thank the Reviewer for pointing out a lack of details with respect to Monte Carlo simulations. We used a nested model built from actual data distributions, without any assumptions. Accordingly, the stimulation did not have hyperparameters typical in machine learning applications, the only external parameter being the number of resamplings (3000 for each draw). We made these modeling choices clearer and expanded this part as follows.

"The effect of movement speed on the distribution of behavioral response types was tested using a nested Monte Carlo simulation framework (Fig. S5). This simulation aimed to model how different movement speeds impact the probability distribution of response types, comparing these simulated outcomes to empirical data. This approach allowed us to determine whether observed differences in response distributions are solely due to speed variations across genotypes or if additional behavioral factors contribute to the differences. First, we calculated the probability of each response type at different specific speed values

(outer model). These probabilities were derived from the grand average of all trials across each genotype, capturing the overall tendency at various speeds. Second, we simulated behavior of virtual flies ($n = 3000$ per genotypes, which falls within the same order of magnitude as the number of experimentally recorded trials from different genotypes) by drawing random velocity values from the empirical velocity distribution specific to the given genotype and then randomly selecting a reaction based on the reaction probabilities associated with the drawn velocity (inner model). Finally, we calculated reaction probabilities for the virtual flies and compared it with real data from animals of the same genotype.

Differences were statistically tested by Chi-squared test.”

(6) The statistical analysis in different experiments needs revisiting. It wasn't clear to me if the authors checked if the data is normally distributed. A simple remedy to this would be to check the normality of data using the Shapiro-Wilk test or Kolmogorov-Smirnov test. Based on the normality check, data should be further analyzed using either parametric or non-parametric statistical tests. Further, the statistical test for the age-dependent behavior response needs revisiting as well. Using two-way ANOVA is not justified given the complexity of the experimental design. Again, after checking for the normality of data, a more rigorous statistical test, such as split-plot ANOVA or a generalized linear model could be used.

We thank the Reviewer for this comment. We performed Kolmogorov-Smirnov test for normality on the data distributions underlying Figure 3, and normality was rejected for all data distributions at $p = 0.05$, which justifies the use of the non-parametric Mann-Whitney U-test. Regarding ANOVA, we would like to point out that the ANOVA hypothesis test design is robust to deviations from normality (Knief and Forstmeier, 2021; Mooi et al., 2018). While the Kruskal-Wallis test is considered a reasonable non-parametric alternative of one-way ANOVA, there is no clear consensus for a non-parametric alternative of two-way ANOVA. Therefore, we left the two-way ANOVA for Figure 5 in place; however, to increase the statistical confidence in our conclusions, we performed Kruskal-Wallis tests for the main effect of age and found significant effects in all genotypes in accordance with the ANOVA, confirming the results (Stop frequency, *DopEcR* $p = 0.0007$; *Dop1R1*, $p = 0.004$; *Dop1R2*, $p = 9.94 \times 10^{-5}$; w^{1118} , $p = 9.89 \times 10^{-13}$; $y^1 w^{67} c^{23}$, $p = 2.54 \times 10^{-5}$; Slowing down frequency, *DopEcR*, $p = 0.0421$; *Dop1R1*, $p = 5.77 \times 10^{-6}$; *Dop1R2*, $p = 0.011$; w^{1118} , $p = 2.62 \times 10^{-5}$; $y^1 w^{67} c^{23}$, $p = 0.0382$; Speeding up frequency, *DopEcR*, $p = 0.0003$; *Dop1R1*, $p = 2.06 \times 10^{-7}$; *Dop1R2*, $p = 2.19 \times 10^{-6}$; w^{1118} , $p = 0.0044$; $y^1 w^{67} c^{23}$, $p = 1.36 \times 10^{-5}$). We also changed the post hoc Tukey-tests to post hoc Mann-Whitney tests in the text to be consistent with the statistical analyses for Figure 3. These resulted in very similar results as the Tukey-tests. Of note, there isn't a straightforward way of correcting for multiple comparisons in this case as opposed to the Tukey's 'honest significance' approach, we thus report uncorrected p values and suggest considering them at $p = 0.01$, which minimizes type I errors. These notes have been added to the 'Data analysis and statistics' Methods section.

(7) The dopamine receptor mutants used in this study are well characterized for learning and memory deficits. In the Parkinson's disease model of Drosophila, there is a loss of DA neurons in specific pockets in the central brain. Hence, it would be apt to use whole animal DA receptor mutants as general DA mutants rather than the Parkinson's disease model. The authors may want to rework the title to reflect the same.

We thank the Reviewer for this comment, which suggests that we were not sufficiently clear on the *Drosophila* lines with DA receptor mutations. We used *Mi{MIC}* random insertion lines for dopamine receptor mutants, namely $y^1 w^{*1}$; *Mi{MIC}Dop1R1^{MI04437}* (BDSC 43773), $y^1 w^{*1}$; *Mi{MIC}Dop1R2^{MI08664}* (BDSC 51098) (Harbison et al., 2019; Pimentel et al., 2016), and w^{1118} ; *PBac{PB}DopEcR^{c02142}/TM6B, Tb¹* (BDSC 10847) (Ishimoto et al., 2013; Petrucci et al., 2020,

2016). These lines carried reported mutations in dopamine receptors, most likely generating partial knock down of the respective receptors. We made this clearer by including the full names at the first occurrence of the lines in Results (beyond those in Methods) and adding references to each of the lines.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) Please think about focusing the manuscript either on the escape response or the PD pathology and provide additional evidence to demonstrate that you indeed have a novel system to address open questions in the field.

As detailed above, we now emphasize more that the main advantage of our single-trial-based approach lies in the appropriate statistical comparison of rich distributions of behavioral data. Please see our response to the ‘Weaknesses’ section for more details.

(2) Please explain the rationale for choosing the genetic lines and provide appropriate genetic controls in the experiments, e.g. trans-heterozygotes. Why use Ddc-Gal4 instead of TH or other specific Split-Gal4 lines?

We thank the Reviewer for this suggestion. We repeated our key experiments with *TH-Gal4* and *NP6510-Gal4* lines. Please see our response to Point #2 of Reviewer #2 for details.

(3) Please proofread the manuscript for omissions. e.g. there's no legend for Fig 4b.

We respectfully point out that the legend is there, and it reads “b, Proportion of a given response type as a function of average fly speed before the shadow presentation. Top, Parkin and α -Syn flies. Bottom, *Dop1R1*, *Dop1R2* and *DopEcR* mutant flies.”

Reviewer #2 (Recommendations For The Authors):

(1) In figure 2(c), representing the average walking speed data for different mutants would be useful to visually correlate the walking differences.

We thank the Reviewer for this suggestion. The average walking speed was added in a scatter plot format, as suggested in the next point of the Reviewer.

(2) The data could be represented more clearly using scatter plots. Also, the color scheme could be more color-blindness friendly.

We thank the Reviewer for this suggestion. We added scatter plots to Fig.2c that indeed represent the distribution of behavioral responses better. We also changed the color scheme and removed red/green labeling.

(3) The manuscript should be checked for typos such as in line 252, 449, 484.

Thank you. We fixed the typos.

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